TACEDGE · PRODUCT DEFINITION PACKAGE

TACEDGE Ground Engineering

V1 · As-Built Product Definition

The evidence-based definition of the V1 production application, prepared for the incoming Interim Development Technical Lead ahead of the V2 technical scoping sprint.

```
STATUS  V1.0 · As-built
CLASS   INTERNAL
SOURCE  TACEDGE Geotech @ 975a2ea · 2026-07-05
INSPECTED 2026-07-17 · static inspection · runtime UNVERIFIED
```

Contents

TACEDGE · V1 PRODUCT DEFINITION · AS-BUILT

00 · Read First

What this package is, how it was produced, and how far to trust it. Read this section before relying on any claim in the sections that follow.

STATUS V1.0 · As-built	CLASS INTERNAL	PRODUCT TACEDGE Ground Engineering V1 ("RockControl")	EVIDENCE SOURCE TACEDGE Geotech @ `975a2ea`
INSPECTION DATE 2026-07-17	AUDIENCE Interim Development Technical Lead · founders		

Purpose

This is an **as-built product definition of TACEDGE Ground Engineering V1** — the production application known in its own repository as RockControl. It was produced for the incoming Interim Development Technical Lead to use during the 30-hour V2 technical scoping sprint. It describes what V1 *is*, not what it should become: it is not a redesign, a V2 architecture recommendation, or a roadmap.

Repository baseline

ITEM	TACEDGE-V1-PD (THIS SITE)	TACEDGE GEOTECH (EVIDENCE SOURCE)
Role	Writable output repository	**Read-only** evidence source
Branch	`claude/tacedge-v1-product-definition-bxrl5e`	`claude/tacedge-v1-product-definition-bxrl5e`
Commit inspected	—	`975a2ea352f1e1ab260c622a66ab59291d8b81b1`
Commit date	—	2026-07-05 (merge of PR #222)

REPOSITORY SAFETY BOUNDARY

tacedge-v1-PD is the only writable repository. TACEDGE Geotech was inspected as a read-only evidence source. No intentional changes were made to the TACEDGE Geotech repository — no files created or edited, no packages installed, no branches, commits, migrations or configuration changes. `git status` was verified clean in TACEDGE Geotech before and after the work.

Evidence methodology

Findings were derived in this order of authority:

1. Current database schema and migrations (`shared/schema*.ts` , `migrations/`).
2. Active routes, pages, components, services and application logic.
3. Tests, fixtures and seed data.
4. Deployment and infrastructure configuration (`fly.toml` , Dockerfile, CI).
5. README and current technical documentation.
6. Git history where needed to explain the present state.
7. Code comments and internal naming.
8. Inference from multiple consistent technical signals.

The V2 Product Definition PDF (in `reference/`) was used for **format and comparison context only**. No V2 claim has been treated as evidence of V1 behaviour.

> THE APPLICATION WAS NOT RUN
>
> V1 could not be run safely during this inspection: it requires a Supabase Postgres database, a session secret and eight external API keys, none of which were available, and installing dependencies or environment files into TACEDGE Geotech was prohibited. **Every runtime-behaviour claim in this package is therefore INFERRED from code or marked UNVERIFIED, and no screenshots of the live product exist here.** Screen descriptions are reconstructed from route registrations, page components and navigation configuration.

Evidence classifications

LABEL	MEANING
IMPLEMENTED	Demonstrably present and functioning in the current codebase.
PARTIAL	Present but incomplete, constrained, inconsistent or not fully connected.
DORMANT	Code, schema or interface exists but does not appear active.
INFERRED	Strongly suggested by the implementation but not directly confirmed.
UNVERIFIED	Could not be validated from the repository or safe access.
NOT IMPLEMENTED	Clearly absent, where absence is relevant.
LEGACY	Retained code, routes, models or assets no longer part of the active product.

Confidence is stated as HIGH / MEDIUM / LOW on registers and sheets. Where confidence is not High, the validation needed is stated.

How to use this site

- Sections 01–02 orient you on the product. Section 03 is the capability register. Sections 04–07 are the functional, screen, design and technical detail. Section 08 maps V1 to the V2 reference. Sections 09–10 hold open questions and appendices.
- Machine-readable evidence lives in `evidence/*.csv` in the tacedge-v1-PD repository — the same data that backs the tables here.
- Diagrams are Mermaid sources in `diagrams/*.mmd` , rendered to SVG by `diagrams/render.sh` — edit the source and re-render.

- Search (top of the sidebar area on built sites) and the capability filters are progressive enhancements; all content works without JavaScript.

How to print the complete package

1. Open /print (also linked from every page as “Print Product Definition”).
2. Use the browser print dialog (Chromium recommended): A4, portrait, default margins off (the stylesheet sets 16–20 mm), background graphics **on**.
3. Save as PDF. The route renders cover, contents and sections 00–10 in order with print-specific styling.

WHAT V1 IS, IN TWO PAGES

01 · Executive Summary

The shortest honest account of the product as built: what it does, who uses it, where it is strong, and where the evidence runs out.

What V1 is

V1 is **RockControl**: a multi-tenant SaaS web application for construction health & safety, geotechnical anchor work and port operations. One Express + React application on Fly.io serves at least four branded tenants through custom domains — Rock Control (rockcontrol.app), TACEDGE (tacedgepm.com), Lyttelton Port Company (lpcreporting.app) and SteepWorks. It is an installable, offline-capable PWA backed by Supabase Postgres, with AI woven through most workflows (Anthropic Claude for analysis and extraction, Deepgram and ElevenLabs for voice).

The primary problem it addresses is the paper-bound field record: drill and grout logs, anchor tests, daily activity sheets, safety forms and toolbox meetings are captured digitally at or near the point of work, reviewed by a project manager, and exported as XLSX, PDF and formal reports for engineers and clients.

The current product model

V1 is **one codebase behaving as three overlapping products**:

- **A geotechnical anchor platform** — projects with zones and sections; anchor designs, grout mixes and drill methods configured per project; AI import of anchor plans from PDF; field drill/grout capture; load testing with AI OCR of paper test sheets; drill-log and test exports; an engineer report builder with revisions.
- **A construction H&S system** — dynamic form templates (incidents, near misses, inspections, asset assessments, JSA sign-ins, Take-5s), safety meetings with recording, transcription and AI analysis, an asset register, and voice-first reporting by phone or in-app.
- **A port-operations skin** — the LPC tenant hides geotech operations entirely and relabels the rest into marine terms with a bespoke dashboard.

Tenancy is data-driven in principle (a `tenant_config` table carries branding, feature flags and copy overrides) but supplemented in practice by hard-coded per-brand branching in both client and server.

Current users and surfaces

Ten database roles collapse into four navigation groups: Management (OrgAdmin, ProjectManager, HSEManager), Supervisor, Field (FieldTech, Subcontractor) and Viewer (ClientViewer). The product has ~40 registered screens across desktop and a phone-first field experience with a bottom navigation bar, offline drafts and a replayable sync queue. A phone-call voice agent (Twilio → ElevenLabs → Claude extraction) files incident and near-miss forms without the app.

What has been demonstrated in live use

The repository shows sustained production activity: 56 applied migrations, a deploy pipeline with migration gating (added after a 2026-05-27 outage), per-tenant seeds for LPC and SteepWorks, and a May 2026 senior-dev cleanup that closed all Critical/High security findings. What cannot be shown from the repository is usage volume, which tenants are commercially active, or the reliability of the offline and voice paths in the field — **this inspection did not run the application**, so all runtime behaviour is inferred from code.

Major limitations

- **Product:** three grouping models coexist (projects, sites, jobs); incidents live in two stores; there is no client/engineer-facing release gate — nothing like a confirmed-only-outward rule exists; six geotech log types advertised in navigation have no implementation.
- **Technical:** no error tracking, APM or metrics (console logs to Fly only); a single shared 1 GB VM; webhook and Twilio validation fail open when secrets are unset; startup env validation is written but never invoked; the 500-line and hook-extraction rules have regressed materially (FacePhotoCanvas is 1,753 lines; 124 inline query/mutation call sites).
- **History:** an entire aviation vertical and a database-trigger "Co-Pilot" event stream were built and dropped within months; their residue (roles, org types, nav entries, columns) remains.

Overall technical posture

Better than its size suggests, worse than its documentation claims. Security fundamentals are genuinely strong for its stage: RLS on every table with a deploy-time audit, hashed reset tokens, strict CSP, per-org repository-layer tenant isolation, and sanitised AI prompt boundaries. The weaknesses are operational (no observability, no staging evidence, one VM) and structural (tenant-specific branching, duplicated status vocabularies, oversized files, stale docs — the repo's own CLAUDE.md is out of date on several counts).

What remains uncertain

Live usage per tenant; offline reliability in the field; voice-call completion rates; AI extraction accuracy; whether the point-cloud worker machine is actually deployed; and which exports are commercially load-bearing. These are catalogued in section 09 as questions for Mike and Dan.

> WHY V1 IS A USEFUL BASELINE FOR THE V2 DECISION
>
> V1 proves the demand-side mechanics the V2 definition depends on: field crews can capture drill/grout/test records digitally, PMs can review and export them, AI can lift paper test sheets and PDF plans into structured data, and multiple real customers have been onboarded onto one codebase. What V1 does *not* prove is the V2 operating spine — configurable work types, a QA/confirm review lifecycle, and a trusted engineer-facing record. The scoping question is therefore not "does the capability exist?" but "does this implementation carry those concepts, or does it resist them?" — and this package gives the Technical Lead the evidence to answer that.

V1 AS IT CURRENTLY EXISTS

02 · Product Brief

The as-built product brief: problem, users, operating model, scope boundary and a reconstructed Definition of Done. Where V1's original intent could not be found in the repository, that is said plainly rather than invented.

WHAT THIS ARTEFACT IS	PRODUCT	ORIGINAL BRIEF	CONFIDENCE
As-built Product Brief, reconstructed from the codebase	RockControl / TACEDGE Ground Engineering V1	None found in the repository	High on scope · Medium on intent

Product problem addressed by V1

Frontline construction and geotechnical work generates records — drill logs, grout logs, anchor tests, sign-ons, incidents, toolbox talks, daily timesheets — that traditionally live on paper and reach the office late, incomplete or not at all. V1 digitises that capture at the point of work, keeps it usable where connectivity is poor, gives managers a live view, and turns the accumulated record into exports and reports a client or engineer can receive.

A second, quieter problem V1 addresses is **multi-customer delivery**: one team operating one codebase for several branded customers with different vocabularies (geotech contractor vs port operator) and different feature needs.

Users and apparent roles

ROLE (DB)	GROUP	WHAT THEY DO IN V1
`FieldTech` / `Subcontractor`	Field	Capture drill/grout logs, fill forms, sign JSAs, record daily activity, report incidents by voice or app.
`SiteSupervisor`	Supervisor	Everything Field does, plus review surfaces and meeting management.
`ProjectManager` / `HSEManager`	Management	Configure projects and setup entities, review and bulk-edit logs, run tests and reports, manage meetings and incidents.
`OrgAdmin`	Management	All of the above plus users, permissions overrides, branding and integrations.
`ClientViewer`	Viewer	Read-only access to projects, testing, documents, submissions.
`Pilot` · `CouncilApprover` · `CouncilAdmin`	—	Aviation-era roles with no remaining surfaces (DORMANT).

Primary user: the field crew member on a phone — the capture surfaces, offline machinery and voice intake all centre on them. **Apparent economic buyer:** the contracting organisation's management (INFERRED from the admin/branding/export emphasis; the repository contains no pricing or billing code at all, so commercial mechanics are outside the codebase).

Current operating model and main workflow

The load-bearing workflow, as built, is: **configure → capture → review → export**. A PM creates a project, defines zones/sections and setup entities (anchor designs, grout mixes, drill methods), and plans anchors manually or by AI PDF import. Field crew work through the plan on the Drill screen; anchor status auto-advances (planned → drilled → grouted → tested). Tests are recorded against standards, often by photographing the paper test sheet for AI OCR. Management reviews on the Drill Summary, flags rows for PM review, and produces XLSX exports, test PDFs, daily activity invoices and built-up engineer reports. Safety capture (forms, meetings, voice reports) runs alongside on the same project spine.

Scope as classified

BAND	CONTENTS
IMPLEMENTED Core	Projects/zones/sections; anchor designs, grout mixes, drill methods; AI anchor-plan import; drill/grout capture with lithology; anchor testing with OCR and exports; Drill Summary; daily activity sheets + DAS specs; dynamic forms + form builder; incidents/near-misses/inspections/assessments; JSA sign-ins; safety meetings with transcription and AI analysis; asset register; voice intake (phone + in-app); report builder with revisions; dashboards; offline PWA; multi-tenant theming; RBAC with per-org overrides.
PARTIAL	Sites and jobs models (semi-legacy, still routed); point-cloud processing (worker not started by the app); e2e testing (config without dependency); notification preferences (consumption untraced); public form-upload flow (auth model unclear).
DORMANT	WebAuthn/passkeys; admin-users page; drone route stubs; startup env validator; components/examples scaffolding.
LEGACY	Aviation vertical (dropped tables; surviving roles/org-types/nav entries); Co-Pilot event stream (dropped); Take-5 hard-coded page; structured voice mode; "Rock Red" design documentation.
NOT IMPLEMENTED (where relevant)	Soil-nail/shotcrete/borehole/instrumentation/slope-monitoring/lab-testing log types (nav placeholders only); QA/confirm review lifecycle; engineer-facing release gate; billing/pricing; observability.

Customer-specific configuration evident in the product

- **LPC (port):** orgType `port` hides Project Operations; marine relabelling; bespoke dashboard; own voice agent, Twilio number and webhook secrets; branding deliberately excluded from the public branding endpoint.
- **SteepWorks:** form-builder flag enabled by migration; meeting recording disabled; "Daily Sign-In" nav re-pointed to a form template; invoice-format DAS export.
- **TACEDGE brand:** marketing site and glass theme, both gated on the literal brand name.
- **Rock Control:** default tenant with geotech opt-in flags enabled.

Reconstructed as-built Definition of Done

> RECONSTRUCTION NOTICE
>
> No original V1 product brief or Definition of Done exists in the repository. The following is **reconstructed from what the implementation visibly optimises for**, and should be validated with the founders rather than treated as agreed history.

1. A field crew member can capture a drill/grout record on a phone, including with degraded connectivity, without paper.
2. A PM can see and correct the drilling record across a project and export it in the formats clients already expect (XLSX, PDF).
3. An anchor test recorded on paper can be digitised by photo and checked for conformance without re-keying.

4. A safety event can be reported in under a minute by voice, from any phone, and reach the right people (Teams/email).
5. A new customer can be onboarded by configuration (branding, flags, copy) rather than a fork — with code-level branching accepted where configuration ran out.

Observable evidence that V1 creates value

- Three-plus production tenants with seeds, custom domains and per-tenant voice numbers.
- Migration cadence of roughly one schema change per week across 2026, tracking real feature demand (zones, DAS specs, job briefs, signatories).
- Deploy-gating added in response to a real outage — the system is operated, not just built.

Success measures that cannot be established from the repository: revenue, retention, active users, capture completion rates, report turnaround times, and paper displacement. None are instrumented.

Principal product limitations

- No review/confirmation lifecycle between capture and client visibility — `needsPmReview` is a flag, not a gate.
- No engineer-facing surface; ClientViewer is a permission level, not a designed experience.
- Work types are hard-coded around anchors; nothing like a work-type configuration exists.
- Structural ambiguity between projects, sites and jobs, and between the two incident stores.
- Customer-specific behaviour has leaked from configuration into code.

Confidence and validation required

Scope claims here are `HIGH` confidence (traceable to code). Intent claims — primary user, buyer, Definition of Done — are `MEDIUM` and marked for founder validation in section 09.

75 CAPABILITIES · EVIDENCE-LED

03 · As-Built Capability Register

Everything V1 can (and pointedly cannot) do, grouped by product domain, each entry classified, evidenced and confidence-rated. Filters are a JavaScript enhancement; the full register always renders.

Capability boundary

WHAT V1 DOES / DOES NOT DO

Does: digital drill, grout and test capture for ground anchors; AI plan import and test-sheet OCR; daily activity and safety form capture; recorded, AI-analysed safety meetings; voice-first incident reporting by phone; multi-tenant branded delivery; XLSX/PDF/report outputs; offline-tolerant field use.

Does not: gate records through a QA/confirm lifecycle before client visibility; serve engineers a designed read-only record; configure new work types from data (anchors are hard-coded; six other log types are nav placeholders); compute pay/invoicing beyond DAS rate exports; provide observability of itself; handle billing.

Authentication

`V1-AUTH-01` · Email + password login `IMPLEMENTED` `CORE`

Users sign in with email and password; sessions are Postgres-backed with a 30-day rolling TTL.

USER / ROLE

All users

INPUTS

Email, password

OUTPUTS

Session cookie (httpOnly, sameSite=lax)

DEPENDENCIES

bcryptjs, express-session, connect-pg-simple

KNOWN LIMITATIONS

Rate-limited 10/15 min. Hard-coded @rockcontrol.app domain check bypasses email verification (server/routes/auth.ts:146).

CONFIDENCE

HIGH

EVIDENCE · V1-AUTH-01

`V1-AUTH-02` · Microsoft OAuth (Azure AD) `IMPLEMENTED` `CORE`

OIDC + PKCE flow against Azure AD with JWKS signature verification; links via oauth_connections.

USER / ROLE

All users

INPUTS

Azure AD identity

OUTPUTS

Session

DEPENDENCIES

jose; MICROSOFT_TENANT_ID/CLIENT_ID/CLIENT_SECRET

KNOWN LIMITATIONS

oauth_provider enum lists google/github but only Microsoft is implemented.

CONFIDENCE

HIGH — validation: Confirm which tenants actively use SSO in production.

EVIDENCE · V1-AUTH-02

`V1-AUTH-03` · WebAuthn / passkeys `DORMANT` `EXPERIMENTAL`

Client and server packages plus a webauthn_credentials table exist, but no WebAuthn endpoints are registered in the route registry.

USER / ROLE

All users

INPUTS

—

OUTPUTS

—

DEPENDENCIES

@simplewebauthn/browser + server

KNOWN LIMITATIONS

Not reachable by users as built.

CONFIDENCE

HIGH — validation: Confirm whether passkeys were ever live or are planned.

EVIDENCE · V1-AUTH-03

V1-AUTH-04 · Password reset + email verification IMPLEMENTED CORE

Reset and verification tokens are 64-char, SHA-256-hashed at rest, single-use, with 1 h / 24 h TTLs; emails sent via Resend.

USER / ROLE

All users

INPUTS

Email

OUTPUTS

Reset/verify emails; updated credentials

DEPENDENCIES

Resend; password_resets, email_verifications tables

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-AUTH-04

V1-AUTH-05 · Access requests + forced password change IMPLEMENTED SUPPORTING

Public request-access form creates review queue rows; accounts flagged mustChangePassword are forced through a change-password screen.

USER / ROLE

Public / new users

INPUTS

Requester details

OUTPUTS

access_requests row; forced password rotation

DEPENDENCIES

access_requests table

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-AUTH-05

Organisations

V1-ORG-01 · Multi-tenant organisations IMPLEMENTED CORE

One Fly app serves multiple custom domains; organizations is the tenant root and every business table carries organization_id, filtered at the repository layer.

USER / ROLE
Platform

INPUTS
Host header / user org

OUTPUTS
Tenant-scoped data access

DEPENDENCIES
organizations table; repository layer

KNOWN LIMITATIONS
No global tenant middleware — isolation depends on every route using loadCurrentUser correctly.

CONFIDENCE
HIGH

EVIDENCE · V1-ORG-01

V1-ORG-02 · Tenant branding and configuration IMPLEMENTED CORE

tenant_config drives brand name, logo, favicon, primary colour, feature flags, copy overrides, support contacts and voice phone; a settings Branding tab uploads assets.

USER / ROLE
OrgAdmin

INPUTS
Branding fields, uploads

OUTPUTS
Per-domain themed UI + PWA manifest

DEPENDENCIES
tenant_config; useTenant hook; routes/manifest.ts

KNOWN LIMITATIONS
PATCH /api/tenant/config has no zod validation; hostname defaults and glass-theme gating are hard-coded per brand.

CONFIDENCE
HIGH

EVIDENCE · V1-ORG-02

`V1-ORG-03` · Per-tenant feature flags and copy overrides `IMPLEMENTED` `CORE`

featureFlags jsonb gates nav items (default-on set vs opt-in set); copyOverrides relabels navigation and titles; orgType=port hides the whole Project Operations group.

USER / ROLE

Platform

INPUTS

tenant_config.featureFlags / copyOverrides

OUTPUTS

Tenant-shaped navigation

DEPENDENCIES

shared/rbac-config.ts NAVIGATION_CONFIG; app-sidebar.tsx

KNOWN LIMITATIONS

Two competing gates for DAS specs (flag vs !isPortOnly). Port relabelling is code-based, bypassing copyOverrides.

CONFIDENCE

HIGH

EVIDENCE · V1-ORG-03

Users & permissions

`V1-USR-01` · User management `IMPLEMENTED` `CORE`

Admins list, create (with invite email), update and deactivate users from Settings.

USER / ROLE

OrgAdmin

INPUTS

User details, role

OUTPUTS

users rows; invite emails

DEPENDENCIES

canManageUsers capability

KNOWN LIMITATIONS

A dormant standalone admin-users.tsx page duplicates this and is not routed.

CONFIDENCE

HIGH

EVIDENCE · V1-USR-01

V1-USR-02 · Role-based access control with per-org overrides IMPLEMENTED CORE

10 roles map to 7 capabilities at none/read/write in a static matrix; per-org overrides stored in role_capability_overrides win over the matrix (cached 30 s).

USER / ROLE

Platform

INPUTS

Role, per-org overrides

OUTPUTS

Effective capabilities per request

DEPENDENCIES

shared/rbac.ts; server/rbac.ts middleware

KNOWN LIMITATIONS

3 aviation-era roles (Pilot, CouncilApprover, CouncilAdmin) remain in the enum and matrix after the module was dropped.

CONFIDENCE

HIGH

EVIDENCE · V1-USR-02

V1-USR-03 · Notification preferences IMPLEMENTED SUPPORTING

Per-user, per-category notification opt-ins stored and editable in Settings.

USER / ROLE

All users

INPUTS

Category toggles

OUTPUTS

notification_preferences rows

DEPENDENCIES

—

KNOWN LIMITATIONS

Consumption side (which sends honour it) not fully traced.

CONFIDENCE

MEDIUM — validation: Confirm which notification paths check preferences.

EVIDENCE · V1-USR-03

Projects

`V1-PRJ-01` · **Projects** IMPLEMENTED CORE

Projects carry code, status (planned/active/complete/on-hold), coordinates, lithology presets, scope jsonb and a test flag; CRUD gated by canManageJobs.

USER / ROLE

PM / OrgAdmin

INPUTS

Project details

OUTPUTS

projects rows

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-PRJ-01

`V1-PRJ-02` · **Zones and sections** IMPLEMENTED CORE

Projects divide into zones (with row direction LTR/RTL and ordering, cloneable) and sections (chainage from/to); zones arrived May 2026 with a "Zone 1" backfill.

USER / ROLE

PM

INPUTS

Zone/section definitions

OUTPUTS

project_zones, project_sections rows

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-PRJ-02

`V1-PRJ-03` · **Project team** IMPLEMENTED SUPPORTING

Named team members (optionally linked to user accounts) are attached to projects and feed DAS crew selection.

USER / ROLE
PM

INPUTS
Member names/roles

OUTPUTS
project_team_members rows

DEPENDENCIES
—

KNOWN LIMITATIONS
userId optional — members may be names only.

CONFIDENCE
HIGH

EVIDENCE · V1-PRJ-03

Documents

`V1-PRJ-04` · **Project documents** IMPLEMENTED SUPPORTING

Per-project file store in a private Supabase bucket with signed 15-minute read URLs.

USER / ROLE
PM / field

INPUTS
Uploaded files

OUTPUTS
project_documents rows + stored objects

DEPENDENCIES
Supabase Storage

KNOWN LIMITATIONS
No offline availability of documents found (client caches API queries, not files).

CONFIDENCE
HIGH — validation: Confirm field usage of project documents offline.

EVIDENCE · V1-PRJ-04

`V1-DOC-01` · **SharePoint document browser** `IMPLEMENTED` `SUPPORTING`

Browse, search and download SharePoint drive contents through Microsoft Graph; project folders sync on demand.

USER / ROLE

Management / viewer

INPUTS

Graph queries

OUTPUTS

Document listings/downloads

DEPENDENCIES

MS Graph; SHAREPOINT_* env

KNOWN LIMITATIONS

Opt-in feature flag; requires per-tenant Azure setup.

CONFIDENCE

HIGH

EVIDENCE · V1-DOC-01

Project setup

`V1-CFG-01` · **Anchor designs** `IMPLEMENTED` `CORE`

Reusable anchor design specs (type, bond/debond lengths, design load) defined per project, optionally per zone/section, inherited by install logs.

USER / ROLE

PM

INPUTS

Design parameters

OUTPUTS

anchor_designs rows

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-CFG-01

`V1-CFG-02` · **Grout mixes and mix types** IMPLEMENTED CORE

Named grout recipes with cement type and water/cement ratio, built on per-project mix-type categories.

USER / ROLE

PM

INPUTS

Mix definitions

OUTPUTS

grout_mixes, grout_mix_types rows

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-CFG-02

`V1-CFG-03` · **Drill methods** IMPLEMENTED CORE

Named drill methods per project for controlled operator selection.

USER / ROLE

PM

INPUTS

Method names

OUTPUTS

drill_methods rows

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-CFG-03

`V1-CFG-04` · Drillers and signatories registries `IMPLEMENTED` `SUPPORTING`

Per-tenant autocomplete registries of driller/crew names and signatories, with last-used ordering.

USER / ROLE

PM / field

INPUTS

Names

OUTPUTS

drillers, signatories rows

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-CFG-04

`V1-PLN-01` · AI anchor-plan import from PDF `IMPLEMENTED` `CORE`

PM uploads a design PDF; Claude Sonnet proposes an anchor plan; the PM reviews, adjusts and applies — no rows are created until approval.

USER / ROLE

PM / OrgAdmin

INPUTS

Design PDF

OUTPUTS

anchor_import_jobs → planned anchor_install_logs

DEPENDENCIES

Anthropic API; Supabase Storage

KNOWN LIMITATIONS

Restricted to OrgAdmin/ProjectManager roles.

CONFIDENCE

HIGH — validation: Accuracy of AI proposals unverified without runtime.

EVIDENCE · V1-PLN-01

V1-PLN-02 · AI job brief extraction IMPLEMENTED SUPPORTING

PDFs are distilled into a structured, versioned job brief that a person must confirm before it is current.

USER / ROLE

PM / supervisor

INPUTS

Source PDFs

OUTPUTS

job_briefs + job_brief_documents rows

DEPENDENCIES

Anthropic API

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-PLN-02

Drill capture

V1-DRL-01 · Drill log capture IMPLEMENTED CORE

Crew record drilling against planned anchors on the /drill grid: reference, method, depths, dates, issues; status auto-advances to drilled.

USER / ROLE

Field crew

INPUTS

Drill parameters per anchor

OUTPUTS

anchor_install_logs updates

DEPENDENCIES

Project setup (designs/methods)

KNOWN LIMITATIONS

Page carries substantial inline logic; offline replay depends on IndexedDB sync queue.

CONFIDENCE

HIGH — validation: End-to-end field behaviour (offline capture, conflict handling) unverified at runtime.

EVIDENCE · V1-DRL-01

V1-DRL-02 · Lithology by depth IMPLEMENTED CORE

Ground profile layers (from/to depth + material) recorded per anchor, using per-project lithology presets.

USER / ROLE

Field crew

INPUTS

Depth intervals + material

OUTPUTS

ground_profile_layers rows

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-DRL-02

V1-DRL-03 · Bulk drill/grout updates and quick-fill IMPLEMENTED SUPPORTING

Bulk PATCH endpoints update many anchors at once; Driller/Grouter quick-fill dialogs prefill repeated values and persist drafts locally.

USER / ROLE

Field crew / supervisor

INPUTS

Bulk selections

OUTPUTS

Bulk anchor updates

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-DRL-03

Grout capture

`V1-GRT-01` · Grout log capture IMPLEMENTED CORE

Grout records against drilled anchors: mix selection from PM presets, volumes, dates; status advances to grouted.

USER / ROLE

Field crew

INPUTS

Grout parameters

OUTPUTS

anchor_install_logs updates

DEPENDENCIES

grout_mixes

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-GRT-01

Testing

`V1-TST-01` · **Anchor test records** IMPLEMENTED CORE

Load-test results per anchor against a selectable standard (EN ISO 22477-5, BS 8081, PTI DC35, NZS 4230, AS 4678), with creep/displacement values, jack asset link and conformance flag.

USER / ROLE

PM / supervisor

INPUTS

Test data + photos

OUTPUTS

anchor_test_results rows

DEPENDENCIES

assets (jack calibration)

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-TST-01

`V1-TST-02` · **Per-load-step dial readings** IMPLEMENTED CORE

Readings per load step across up to three dial gauges.

USER / ROLE

PM / supervisor

INPUTS

Dial readings

OUTPUTS

anchor_test_readings rows

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-TST-02

V1-TST-03 · AI test-sheet photo OCR + conformance analysis IMPLEMENTED CORE

A photo of a paper test sheet is OCR-extracted by Claude into readings; a second AI pass analyses load/displacement conformance. Human review precedes saving.

USER / ROLE

PM / supervisor

INPUTS

Test-sheet photo

OUTPUTS

Proposed readings + conformance assessment

DEPENDENCIES

Anthropic API

KNOWN LIMITATIONS

Extraction accuracy unverified without runtime.

CONFIDENCE

HIGH — validation: Validate OCR accuracy rates with real test sheets.

EVIDENCE · V1-TST-03

V1-TST-04 · Test exports (XLSX + engineering PDF) IMPLEMENTED CORE

Single-test XLSX export, bulk XLSX, and an engineering test-report PDF.

USER / ROLE

PM

INPUTS

Test results

OUTPUTS

XLSX / PDF files

DEPENDENCIES

xlsx; pdfkit

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-TST-04

Daily records

`V1-DAS-01` · **Daily activity sheets** IMPLEMENTED CORE

Shift logs of crew and plant hours captured as daily-activity forms, with per-project crew/plant selection.

USER / ROLE

Field / supervisor

INPUTS

Crew hours, plant hours, shift details

OUTPUTS

forms rows (type=daily-activity)

DEPENDENCIES

DAS specs

KNOWN LIMITATIONS

Largest page in the codebase; DAS gating differs between flag and port checks.

CONFIDENCE

HIGH

EVIDENCE · V1-DAS-01

`V1-DAS-02` · **DAS specs (positions, personnel, plant, rates)** IMPLEMENTED CORE

Org-level catalogues of positions with charge rates, home bases, personnel and plant with day/hour/standby rates, assigned per project.

USER / ROLE

OrgAdmin / PM

INPUTS

Rate/roster definitions

OUTPUTS

7 das_* tables

DEPENDENCIES

—

KNOWN LIMITATIONS

Recent addition (June 2026) — production usage unverified.

CONFIDENCE

HIGH — validation: Confirm DAS specs are in live use and rates are maintained.

EVIDENCE · V1-DAS-02

`V1-DAS-03` · **DAS PDF and invoice XLSX export** `IMPLEMENTED` `CORE`

Daily activity sheets export to PDF (single and batch) and an invoice-format XLSX.

USER / ROLE

PM

INPUTS

DAS records

OUTPUTS

PDF / XLSX files

DEPENDENCIES

pdfkit; xlsx

KNOWN LIMITATIONS

SteepWorks-specific invoice format noted in TimesheetDAS component.

CONFIDENCE

HIGH

EVIDENCE · V1-DAS-03

Safety

`V1-SAF-01` · **Incident reports** `IMPLEMENTED` `CORE`

Incidents captured via form templates (web or voice), listed and managed with severity/status; AI analysis proposes summary and actions; optional SharePoint sync and Teams alert.

USER / ROLE

All safety roles

INPUTS

Incident details

OUTPUTS

forms rows (incident-report); incidents table; SharePoint items; Teams cards

DEPENDENCIES

form_templates; Anthropic; MS Graph (optional)

KNOWN LIMITATIONS

Two overlapping stores: a dedicated incidents table and form-based incident-reports.

CONFIDENCE

HIGH — validation: Clarify which incident store is authoritative in production.

EVIDENCE · V1-SAF-01

V1-SAF-02 · Near-miss reports IMPLEMENTED CORE

Near-miss forms captured, listed and (for voice submissions) alerting via Teams.

USER / ROLE

All safety roles

INPUTS

Near-miss details

OUTPUTS

forms rows (near-miss)

DEPENDENCIES

form_templates

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-SAF-02

V1-SAF-03 · Site inspections + asset assessments IMPLEMENTED CORE

Template-driven inspection and plant-assessment forms with list views.

USER / ROLE

Management / supervisor

INPUTS

Inspection/assessment details

OUTPUTS

forms rows

DEPENDENCIES

form_templates

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-SAF-03

`V1-SAF-04` · JSA daily sign-ins `IMPLEMENTED` `CORE`

Daily sign-ins recorded against a parent JSA form (parentFormId), with signatures; SteepWorks re-points its "Daily Sign-In" nav item at a form template.

USER / ROLE

Field crew

INPUTS

Signature + acknowledgement

OUTPUTS

Child forms rows

DEPENDENCIES

form_templates.dailySigninEnabled

KNOWN LIMITATIONS

No evidence of sign-on gating drilling (V2 SOS-4 concept is not present in V1).

CONFIDENCE

HIGH

EVIDENCE · V1-SAF-04

`V1-AST-01` · Asset register with certification expiry `IMPLEMENTED` `CORE`

Register of jacks, gauges, vehicles, plant and safety equipment with certification dates, calibration values and an expiring-assets view; jack assets link to anchor tests.

USER / ROLE

Management

INPUTS

Asset details, cert dates

OUTPUTS

assets rows

DEPENDENCIES

—

KNOWN LIMITATIONS

Category/status are free text.

CONFIDENCE

HIGH

EVIDENCE · V1-AST-01

Meetings

V1-MTG-01 · Safety meetings / toolbox talks IMPLEMENTED CORE

Meetings of 7 types with scheduling, status lifecycle, participants with signatures, action tasks and photos.

USER / ROLE

All roles

INPUTS

Meeting details, attendance

OUTPUTS

safety_meetings + 3 child tables

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-MTG-01

V1-MTG-02 · Meeting recording and live transcription IMPLEMENTED CORE

Browser recording with Deepgram streaming transcription (or an OpenAI Realtime session), audio upload, and transcript storage; recordings buffered in IndexedDB.

USER / ROLE

Host

INPUTS

Microphone audio

OUTPUTS

Audio file + transcript jsonb

DEPENDENCIES

Deepgram; OpenAI Realtime; recording-idb.ts

KNOWN LIMITATIONS

SteepWorks disables recording via meetingVoiceRecording=false.

CONFIDENCE

HIGH — validation: Realtime path vs Deepgram path usage split unverified.

EVIDENCE · V1-MTG-02

V1-MTG-03 · AI meeting analysis and Q&A IMPLEMENTED CORE

Claude analyses the transcript into a structured summary; an ask-AI endpoint answers questions about the meeting.

USER / ROLE
Management

INPUTS
Transcript

OUTPUTS
aiAnalysis jsonb

DEPENDENCIES
Anthropic API

KNOWN LIMITATIONS
None noted.

CONFIDENCE
HIGH

EVIDENCE · V1-MTG-03

V1-MTG-04 · Meeting export IMPLEMENTED SUPPORTING

Meeting record exports to a document (PDF).

USER / ROLE
Management

INPUTS
Meeting record

OUTPUTS
Export file

DEPENDENCIES
—

KNOWN LIMITATIONS
None noted.

CONFIDENCE
MEDIUM

EVIDENCE · V1-MTG-04

Forms

`V1-FRM-01` · **Form templates + form builder** IMPLEMENTED CORE

Custom form templates (schema jsonb, 7 field types) authored in a drag-and-drop builder; 11 form types supported.

USER / ROLE

Management

INPUTS

Template definitions

OUTPUTS

form_templates rows

DEPENDENCIES

@dnd-kit

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-FRM-01

`V1-FRM-02` · **Dynamic form fill and submission** IMPLEMENTED CORE

Forms render dynamically from template schema and submit with attachments; drafts survive offline via IndexedDB.

USER / ROLE

Field crew

INPUTS

Form data + attachments

OUTPUTS

forms rows

DEPENDENCIES

dynamic-form-renderer

KNOWN LIMITATIONS

formData is unstructured jsonb — shape varies by template.

CONFIDENCE

HIGH

EVIDENCE · V1-FRM-02

`V1-FRM-03` · Submissions inbox IMPLEMENTED CORE

Submitted forms listed with status lifecycle (draft → pending → approved/rejected → completed → archived), detail dialog and archive filter.

USER / ROLE

All roles

INPUTS

Status changes

OUTPUTS

forms status updates

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-FRM-03

`V1-VAR-01` · Contract variations IMPLEMENTED CORE

Variations with number, reason (ground-conditions/design-change/client-request), status workflow (draft → submitted → approved/rejected) and Teams notification.

USER / ROLE

Management / supervisor

INPUTS

Variation details

OUTPUTS

variations rows

DEPENDENCIES

—

KNOWN LIMITATIONS

Opt-in feature flag.

CONFIDENCE

HIGH

EVIDENCE · V1-VAR-01

Voice

`V1-VCE-01` · **Phone voice intake (Twilio + ElevenLabs)** IMPLEMENTED CORE

A caller dials a per-tenant number; Twilio bridges audio over WebSocket to an ElevenLabs conversational agent that gathers a report and submits it via server tools; on hang-up the narrative is extracted into a structured form.

USER / ROLE

Field callers

INPUTS

Phone call audio

OUTPUTS

forms rows (source=voice) + notifications

DEPENDENCIES

Twilio; ElevenLabs; Anthropic (extraction); HMAC stream token

KNOWN LIMITATIONS

Per-tenant config is env-var-driven with LPC-specific variables and a hard-coded fallback agent ID; webhook validation fails open when secrets are unset.

CONFIDENCE

HIGH — validation: Live call quality and completion rates unverifiable statically.

EVIDENCE · V1-VCE-01

V1-VCE-02 · In-app voice reporting IMPLEMENTED CORE

A Voice Report dialog transcribes speech in the browser via a short-lived Deepgram key and submits the narrative through the same extraction pipeline.

USER / ROLE

All users

INPUTS

Microphone audio

OUTPUTS

forms rows (source=voice)

DEPENDENCIES

Deepgram; Anthropic

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-VCE-02

V1-VCE-03 · AI narrative-to-form extraction IMPLEMENTED CORE

Claude Haiku turns a spoken narrative into structured formData with confidence scores and a missing-required list; a legacy structured mode is retained for back-compatibility.

USER / ROLE

Platform

INPUTS

Transcript

OUTPUTS

Structured formData + extraction metadata

DEPENDENCIES

Anthropic API

KNOWN LIMITATIONS

Legacy structured path still in production (voice.ts:359).

CONFIDENCE

HIGH

EVIDENCE · V1-VCE-03

Maps & spatial

`V1-MAP-01` · Project map with NZ basemaps IMPLEMENTED CORE

MapLibre map with LINZ aerial/cadastral basemaps (client API keys), Mapbox fallback, project locations and annotation drawing (terra-draw).

USER / ROLE

Management

INPUTS

Project coordinates; drawings

OUTPUTS

map_annotations GeoJSON

DEPENDENCIES

maplibre-gl; LINZ keys

KNOWN LIMITATIONS

Default centre hard-coded to Christchurch; LPC centre hard-coded.

CONFIDENCE

HIGH

EVIDENCE · V1-MAP-01

`V1-MAP-02` · Face-photo anchor map IMPLEMENTED CORE

Rock-face photos carry pinned anchors, annotations and a bi-axial scale so anchor positions live on imagery rather than coordinates.

USER / ROLE

PM / field

INPUTS

Face photos; pin placements

OUTPUTS

face_photos rows; anchor facePhotoId links

DEPENDENCIES

—

KNOWN LIMITATIONS

FacePhotoCanvas has regrown well past its documented post-refactor size — fragile area.

CONFIDENCE

HIGH

EVIDENCE · V1-MAP-02

`V1-MAP-03` · Map AI chat `IMPLEMENTED` `EXPERIMENTAL`

A per-project map chat endpoint sends map context to Claude Sonnet for spatial Q&A.

USER / ROLE
Management

INPUTS
Chat + map context

OUTPUTS
AI responses

DEPENDENCIES
Anthropic API

KNOWN LIMITATIONS
Usage in production unknown.

CONFIDENCE
MEDIUM — validation: Confirm whether map AI chat is used.

EVIDENCE · V1-MAP-03

`V1-MAP-04` · Terrain elevation profiles `IMPLEMENTED` `SUPPORTING`

Elevation profiles and stats computed via the keyless Open-Meteo elevation API.

USER / ROLE
Management

INPUTS
Line/point geometry

OUTPUTS
Elevation data

DEPENDENCIES
Open-Meteo

KNOWN LIMITATIONS
None noted.

CONFIDENCE
MEDIUM

EVIDENCE · V1-MAP-04

Drone & survey

V1-DRN-01 · Drone surveys and files IMPLEMENTED SUPPORTING

Drone surveys per project with image and point-cloud files, thumbnails, annotations and downloads.

USER / ROLE
Management

INPUTS
Uploaded survey files

OUTPUTS
drone_surveys, drone_survey_files rows

DEPENDENCIES
Supabase Storage

KNOWN LIMITATIONS
None noted.

CONFIDENCE
HIGH

EVIDENCE · V1-DRN-01

V1-DRN-02 · AI rock-face / satellite analysis IMPLEMENTED EXPERIMENTAL

Claude Sonnet analyses drone imagery (rock face, drainage, slope stability, anchor planning), compares surveys, answers follow-ups and drafts work orders.

USER / ROLE
Management

INPUTS
Imagery + prompts

OUTPUTS
analysisResults jsonb; work orders

DEPENDENCIES
Anthropic API

KNOWN LIMITATIONS
Engineering reliability of AI analysis is unestablished.

CONFIDENCE
HIGH — validation: Confirm whether AI survey analysis is used commercially or exploratory.

EVIDENCE · V1-DRN-02

`V1-DRN-03` · **Point-cloud processing (Potree + FACETS)** PARTIAL EXPERIMENTAL

LAS/LAZ files queue to a pg-boss worker on a separate machine that runs PotreeConverter and CloudCompare FACETS, publishing web-viewable octrees; a three/potree viewer renders them.

USER / ROLE

Management

INPUTS

LAS/LAZ files

OUTPUTS

Octree tiles; facets JSON

DEPENDENCIES

pg-boss; PotreeConverter; CloudCompare; separate Fly machine

KNOWN LIMITATIONS

Worker is not started by the main app; whether a worker machine is deployed is unverifiable from the repo.

CONFIDENCE

MEDIUM — validation: Confirm the worker machine exists in production and jobs complete.

EVIDENCE · V1-DRN-03

Reporting

`V1-RPT-01` · **Report builder with revisions** IMPLEMENTED CORE

Project reports authored in a Tiptap block editor from skeleton templates, with a text state machine and immutable revision snapshots carrying prepared/reviewed/authorised signatories.

USER / ROLE

Management

INPUTS

Report content; project data bundle

OUTPUTS

reports + report_revisions rows

DEPENDENCIES

@tiptap; @react-pdf/renderer

KNOWN LIMITATIONS

Status model is route-enforced text, not a DB enum.

CONFIDENCE

HIGH

EVIDENCE · V1-RPT-01

`V1-RPT-02` · AI report proposals `IMPLEMENTED` `SUPPORTING`

Claude Sonnet (tool-use) proposes report sections from the project data bundle; a person accepts or edits.

USER / ROLE

Management

INPUTS

Project data bundle

OUTPUTS

Proposed report content

DEPENDENCIES

Anthropic API

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-RPT-02

`V1-RPT-03` · Drill-log report and XLSX exports `IMPLEMENTED` `CORE`

Per-project drill-log report plus XLSX exports built from a Python-generated template, and a bulk email-to-teammate flow.

USER / ROLE

Management

INPUTS

Anchor logs

OUTPUTS

XLSX files; emails

DEPENDENCIES

xlsx; openpyxl templates; Resend

KNOWN LIMITATIONS

Email recipients restricted to tenant teammates.

CONFIDENCE

HIGH

EVIDENCE · V1-RPT-03

Dashboards

`V1-DSH-01` · **Role-adaptive dashboard** IMPLEMENTED CORE

Dashboard widgets vary by role group; management overview aggregates stats via SQL count-filter queries.

USER / ROLE

All roles

INPUTS

—

OUTPUTS

Overview stats

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-DSH-01

`V1-DSH-02` · **LPC port dashboard** IMPLEMENTED CUSTOMER-SPECIFIC

A bespoke port dashboard with its own navy palette and port KPIs, rendered for the LPC tenant.

USER / ROLE

LPC users

INPUTS

—

OUTPUTS

Port KPI view

DEPENDENCIES

—

KNOWN LIMITATIONS

Entirely customer-specific; bypasses the token system.

CONFIDENCE

HIGH

EVIDENCE · V1-DSH-02

`V1-SRCH-01` · **Global search** `IMPLEMENTED` `SUPPORTING`

A global /api/search endpoint feeds a command-palette search UI.

USER / ROLE

All roles

INPUTS

Query string

OUTPUTS

Cross-entity results

DEPENDENCIES

—

KNOWN LIMITATIONS

Coverage of entities not fully traced.

CONFIDENCE

MEDIUM

EVIDENCE · V1-SRCH-01

AI assistant

`V1-AIA-01` · **"Ask RC" AI assistant** `IMPLEMENTED` `SUPPORTING`

A streaming chat assistant (Claude Haiku tool loop) answers questions using read-only tenant-scoped tools.

USER / ROLE

All roles

INPUTS

Chat messages

OUTPUTS

Streamed answers

DEPENDENCIES

Anthropic API; server/services/assistant-tools.ts (760 lines)

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH — validation: Adoption unknown.

EVIDENCE · V1-AIA-01

Calendar

`V1-CAL-01` · **Outlook-synced calendar** IMPLEMENTED SUPPORTING

Calendar events with optional Outlook sync via delegated Microsoft OAuth (separate consent flow).

USER / ROLE
Management / viewer

INPUTS
Events; OAuth consent

OUTPUTS
calendar_events rows; Outlook events

DEPENDENCIES
MS Graph Calendars.ReadWrite

KNOWN LIMITATIONS
Port tenant relabels this as vessel/maintenance scheduling.

CONFIDENCE
HIGH

EVIDENCE · V1-CAL-01

Notes

`V1-NOT-01` · **Project notes** IMPLEMENTED SUPPORTING

Markdown notes scoped to projects with attachments, visibility and pinning.

USER / ROLE
Field / supervisor

INPUTS
Note content

OUTPUTS
notes rows

DEPENDENCIES
—

KNOWN LIMITATIONS
Some attachments stored as inline data URLs in jsonb.

CONFIDENCE
HIGH

EVIDENCE · V1-NOT-01

Integrations

V1-INT-01 · SharePoint incident sync IMPLEMENTED SUPPORTING

Incident and analysis records sync one-way into SharePoint lists per tenant config.

USER / ROLE
Platform

INPUTS
Incident/analysis records

OUTPUTS
SharePoint list items

DEPENDENCIES
MS Graph; sharepoint_config

KNOWN LIMITATIONS
Duplicate modules (sharepoint.ts vs sharepoint-service.ts) — residual debt.

CONFIDENCE
HIGH — validation: Confirm which tenants have sync enabled.

EVIDENCE · V1-INT-01

V1-INT-02 · Teams notifications IMPLEMENTED SUPPORTING

Teams webhook cards fire for variations and voice-submitted incidents/near-misses.

USER / ROLE
Platform

INPUTS
Events

OUTPUTS
Teams cards

DEPENDENCIES
TEAMS_WEBHOOK_URL

KNOWN LIMITATIONS
Single webhook URL env — not per-tenant in tenant_config.

CONFIDENCE
HIGH

EVIDENCE · V1-INT-02

`V1-INT-03` · **Transactional + notification email** `IMPLEMENTED` `CORE`

Resend sends verification, reset, invite and notification emails with per-brand from-addresses.

USER / ROLE

Platform

INPUTS

Events

OUTPUTS

Emails

DEPENDENCIES

RESEND_API_KEY

KNOWN LIMITATIONS

Per-brand from-addresses hard-coded in code.

CONFIDENCE

HIGH

EVIDENCE · V1-INT-03

Offline & PWA

`V1-OFF-01` · **Offline-first field capture (PWA)** `IMPLEMENTED` `CORE`

Service worker pre-caches key API responses; IndexedDB holds a replayable mutation sync-queue (idempotency IDs, retry/backoff), a React Query cache and form drafts; per-tenant PWA manifest served by host.

USER / ROLE

Field crew

INPUTS

Offline mutations

OUTPUTS

Replayed API calls on reconnect

DEPENDENCIES

sw.js; idb; routes/manifest.ts

KNOWN LIMITATIONS

Depth of offline coverage (which mutations enqueue) not exhaustively traced; behaviour unverified at runtime.

CONFIDENCE

MEDIUM — validation: Field-test a full offline shift and resync.

EVIDENCE · V1-OFF-01

Legacy

`V1-LEG-01` · **Sites model (pre-project)** PARTIAL LEGACY

Sites with contacts and files remain routed and navigable but the geotech operating model has moved to projects/zones.

USER / ROLE

Management

INPUTS

Site details

OUTPUTS

sites rows

DEPENDENCIES

—

KNOWN LIMITATIONS

Overlapping grouping units: sites, jobs and projects coexist.

CONFIDENCE

HIGH — validation: Confirm whether any tenant still works from sites.

EVIDENCE · V1-LEG-01

`V1-LEG-02` · **Jobs model (H&S grouping)** PARTIAL LEGACY

Jobs with codes, membership and progress remain active for H&S flows and job lookup by code.

USER / ROLE

Field / management

INPUTS

Job details

OUTPUTS

jobs rows

DEPENDENCIES

—

KNOWN LIMITATIONS

Duplicates project semantics for geotech tenants.

CONFIDENCE

HIGH — validation: Clarify intended future of jobs vs projects.

EVIDENCE · V1-LEG-02

V1-LEG-03 · Aviation module remnants LEGACY LEGACY

A full aviation/council-consent vertical (22 tables, 7 enums, 10 nav entries) was built (migrations 0005–0012) and dropped (0055). Nav config entries, 3 roles and 2 org types remain.

USER / ROLE

—

INPUTS

—

OUTPUTS

—

DEPENDENCIES

—

KNOWN LIMITATIONS

Enum values and nav config cannot be reached but add cognitive load.

CONFIDENCE

HIGH

EVIDENCE · V1-LEG-03

V1-LEG-04 · Co-Pilot event stream LEGACY LEGACY

A database-trigger event stream ("brain") was created (0047–0049) and dropped (0055); the safety_meetings.planned_work column survives.

USER / ROLE

—

INPUTS

—

OUTPUTS

—

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

CONFIDENCE

HIGH

EVIDENCE · V1-LEG-04

`V1-LEG-05` · **Placeholder geotech log types** NOT IMPLEMENTED EXPERIMENTAL

Six nav entries (soil nails, shotcrete, boreholes, instrumentation, slope monitoring, lab testing) exist behind opt-in flags with no route components.

USER / ROLE

—

INPUTS

—

OUTPUTS

—

DEPENDENCIES

—

KNOWN LIMITATIONS

Would render NotFound if a flag were enabled.

CONFIDENCE

HIGH

EVIDENCE · V1-LEG-05

Marketing

`V1-MKT-01` · **TacEdge marketing site inside the SPA** IMPLEMENTED CUSTOMER-SPECIFIC

Eight marketing pages plus a landing page ship inside the product bundle and render only when brandName === "TacEdge".

USER / ROLE

Public

INPUTS

—

OUTPUTS

Contact/trial submissions

DEPENDENCIES

—

KNOWN LIMITATIONS

Marketing content code-splits but ships in every tenant deployment.

CONFIDENCE

HIGH

EVIDENCE · V1-MKT-01

HOW V1 BEHAVES AS AN OPERATIONAL SYSTEM

04 · Functional Architecture

Roles, surfaces, workflows, lifecycles and data flows — of V1 as built. V2 concepts are not used to fill gaps; where V1 has no equivalent of a V2 behaviour, that absence is stated.

System context

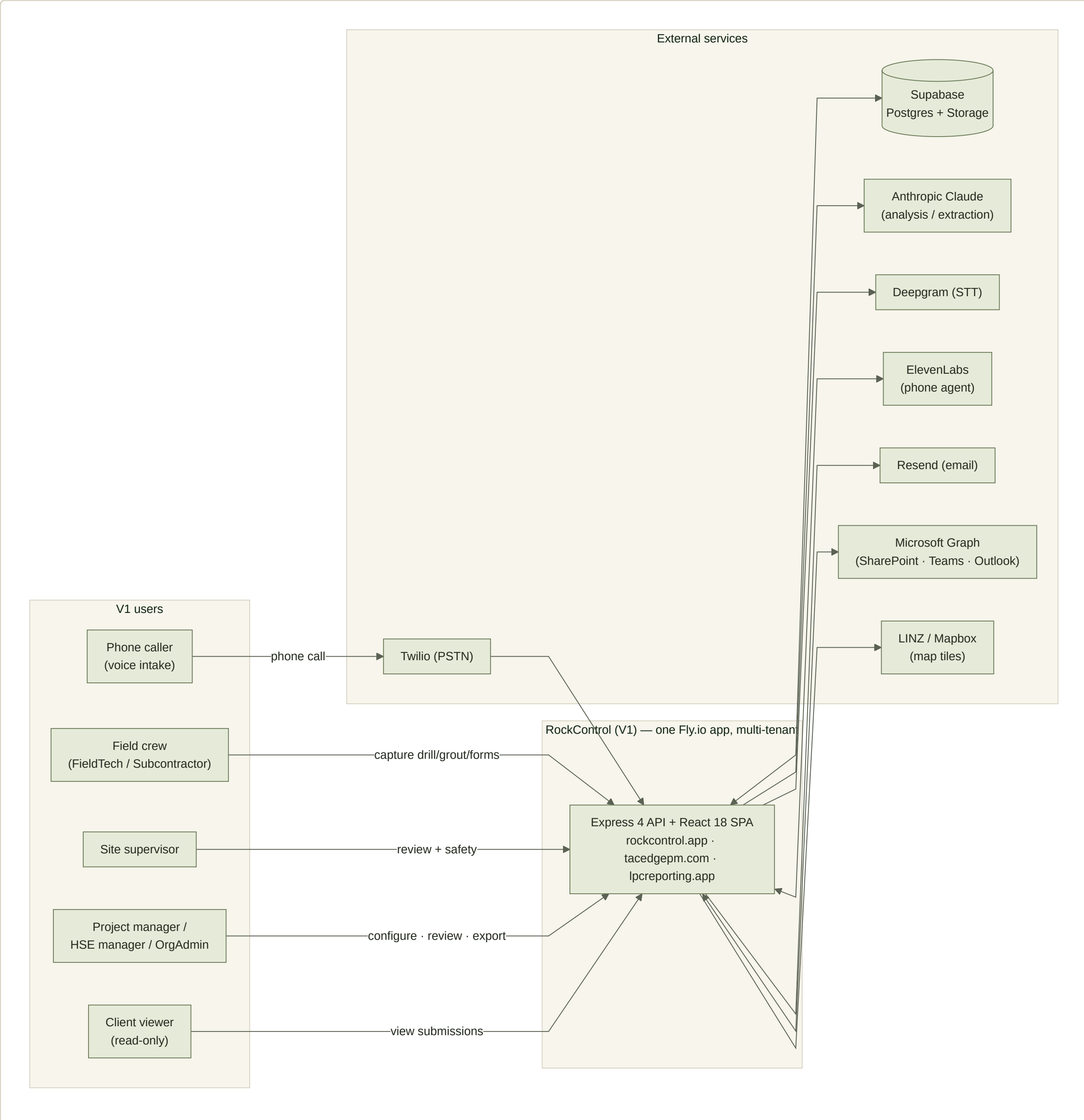

FIG-04-01 V1 system context. One multi-tenant application; eight external service dependencies. Source: diagrams/d01-system-context.mmd.

User roles and permission boundaries

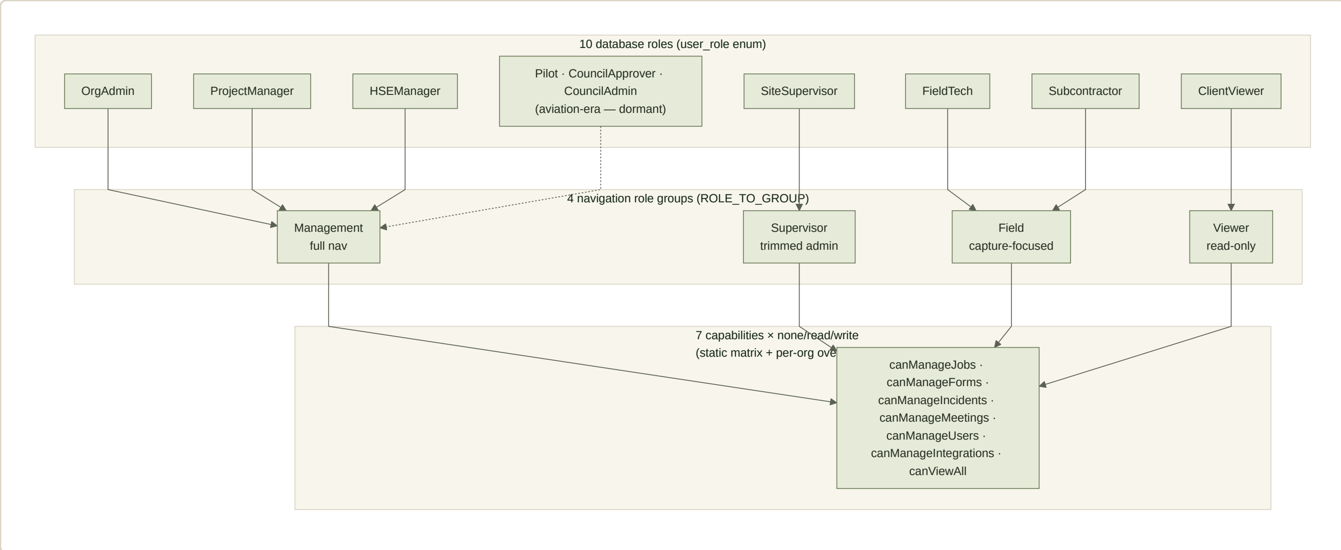

FIG-04-02 Role model: 10 DB roles → 4 nav groups → 7 capabilities at none/read/write, with per-org overrides. Source: diagrams/d02-user-role-map.mmd.

- Permission enforcement is **per-route middleware** (requireCapability / requireRoles) — there is no global RBAC layer. The router itself gates almost nothing; navigation visibility does the front-line shaping.
- Scoping: ClientViewer and Subcontractor are job/global scoped; PM, SiteSupervisor and FieldTech are project-membership scoped; OrgAdmin and HSEManager see org-wide (server/rbac.ts).
- Tenant isolation: every repository query filters `organizationId` taken from the session user — never from the request body.

Product surfaces and navigation

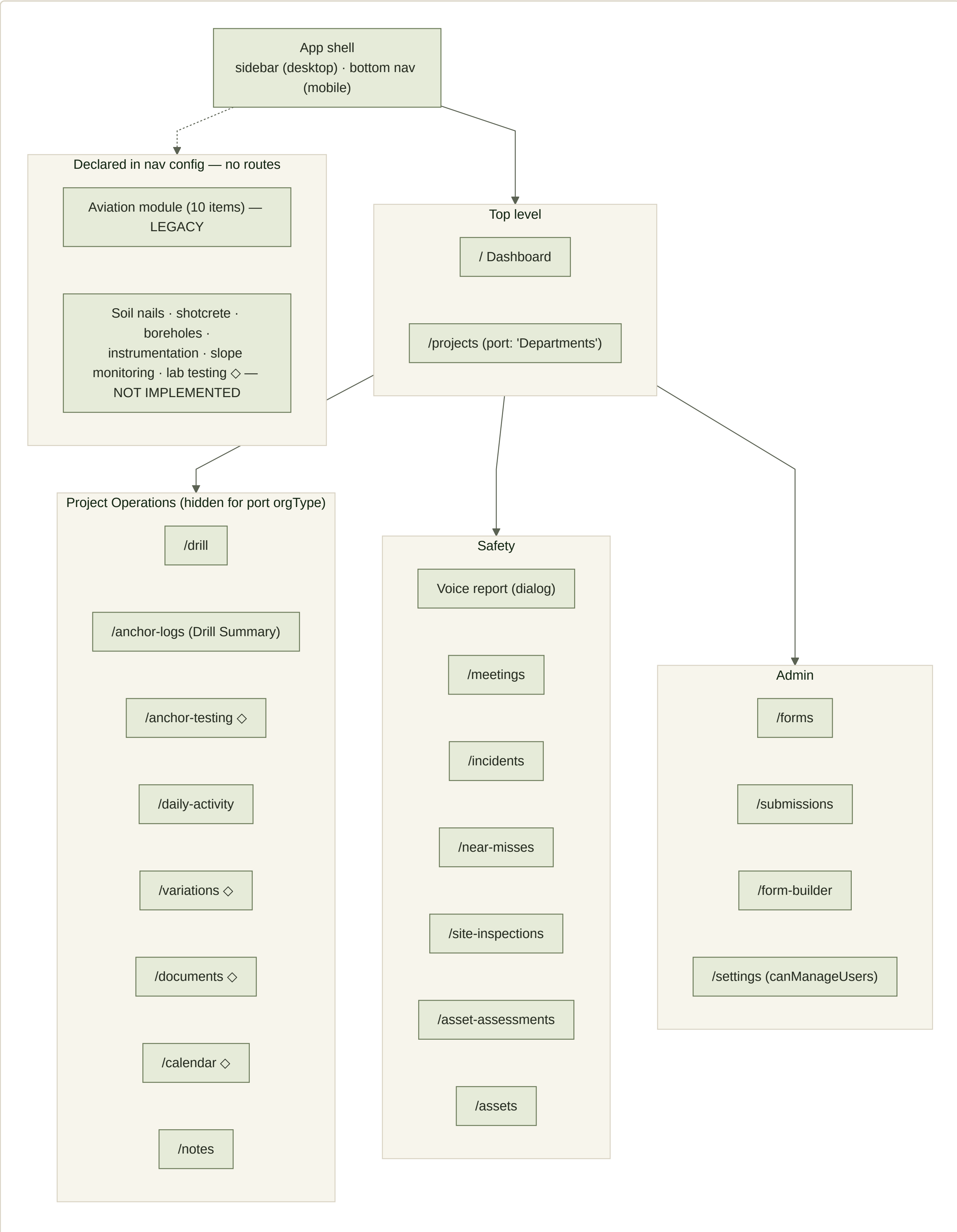

FIG-04-03 Navigation architecture. ◇ marks opt-in feature-flag items. The dormant Aviation group and six unimplemented log types remain declared in config. Source: diagrams/d04-navigation.mmd.

- **Desktop:** pinned sidebar navigation driven by NAVIGATION_CONFIG (role group × module × orgType × feature flags × capabilities), with tenant copy overrides.
- **Mobile:** a fixed bottom bar — geotech tenants get Home / This Page / brand / Drill / Voice; the port tenant gets a legacy role-driven four-button layout. A "This Page" sheet exposes per-page actions.
- **Voice:** a sidebar item is a sentinel (`__voice__`) that opens the in-app voice dialog; the phone agent needs no app at all.

End-to-end workflow

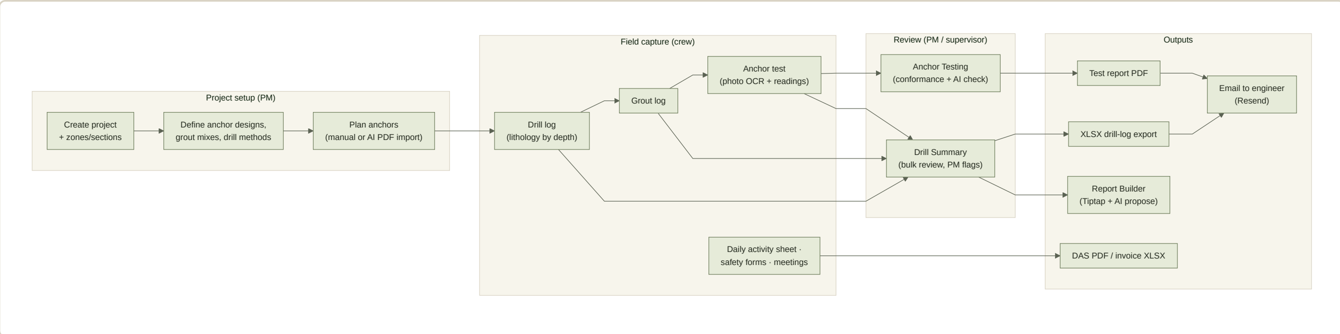

FIG-04-04 The load-bearing V1 workflow: configure → capture → review → export. Source: diagrams/d03-primary-workflow.mmd.

Project lifecycle

`planned → active → complete` with `on-hold` off-path (project_status enum). Projects are soft-deleted (`deletedAt`). A project's setup entities and plan usually precede field work, but nothing enforces ordering.

Work-item (anchor) lifecycle

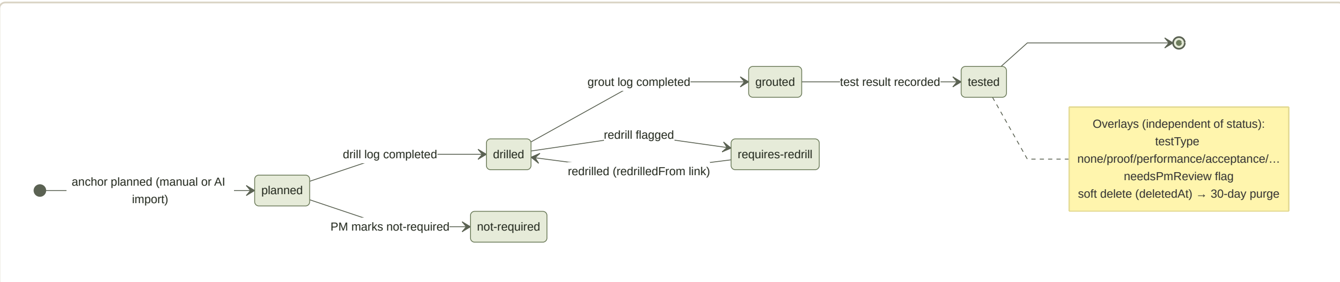

FIG-04-05 Anchor status model (anchor_status enum). Status auto-advances from field actions; requires-redrill links a replacement via redrilledFrom. Source: diagrams/d05-anchor-lifecycle.mmd.

> WHAT V1 DOES NOT HAVE
>
> There is **no review-state dimension** (in-progress / submitted / confirmed / re-opened) and no confirmed-only-outward rule. The nearest equivalents are the `needsPmReview` boolean on anchor logs and the form status lifecycle (draft → pending → approved/rejected → completed → archived), which applies to form submissions, not anchors. Editability is governed by capabilities, not record state.
>
> NOT IMPLEMENTED

Record lifecycles and ownership

- **Forms:** draft → pending → approved / rejected → completed → archived; source is web or voice; JSA sign-ins chain to a parent form.

- **Reports:** a text state machine at the route layer with immutable revision snapshots carrying prepared/reviewed/authorised names.
- **Meetings:** scheduled → recording → processing → completed / cancelled.
- **Ownership:** creator IDs are recorded throughout, but editing rights come from role capabilities — any write-capable user in the org (within project scoping) can edit most records. There is no single-owner-per-state rule.
- **Soft delete:** pervasive `deletedAt` columns; archived anchor logs hard-purge after 30 days via the in-process worker.

Main data flows — capture to report

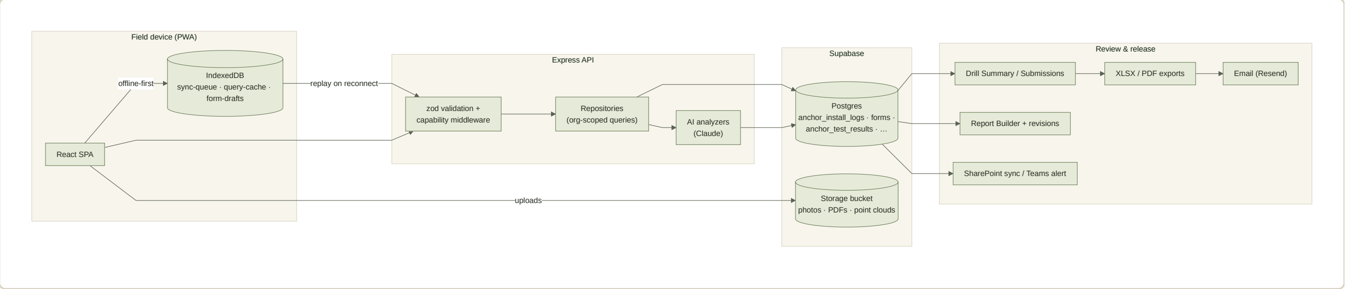

FIG-04-06 Capture-to-report flow, including the offline replay path and outbound integrations. Source: diagrams/d06-capture-to-report.mmd.

Exceptions and error paths

- Field exceptions are operator-declared: issue flags on anchor logs, requires-redrill status, incident/near-miss forms. Voice-submitted incidents and near misses additionally fire Teams notifications.
- Offline mutations queue in IndexedDB with idempotency IDs; replay retries 5xx up to five times, drops other 4xx, and treats 409 as an already-applied duplicate.
- Server errors return `errMessage(err)` with a 500; the client toasts. There is no user-facing error tracking beyond logs.

Notifications and alerts

- Teams webhook cards: variations, voice incidents/near-misses.
- Email (Resend): auth flows, invites, log-set emails, port voice notifications (`PORT_VOICE_NOTIFICATION_EMAILS`).
- Per-user notification_preferences exist; how consistently senders consult them is untraced (PARTIAL).
- No push notifications, in-app inbox, or SMS.

Mobile, desktop and offline behaviour

- Layout is locked to dynamic viewport height with body scroll containment for field stability; pull-to-refresh wraps content.
- Service worker pre-caches form templates, projects, sites and tenant config; a stale-chunk self-heal reloads once on failed lazy imports.
- Offline: drafts and query cache in IndexedDB; sync queue replays on the `online` event. Coverage depth by mutation type is not exhaustively traced. INFERRED

Customer-specific logic (functional view)

- **Port (LPC):** Project Operations hidden; "Departments", "Shift Logs", marine labels; bespoke dashboard; port bottom nav; LPC voice tenant.

- **SteepWorks:** recording disabled; daily sign-in re-pointed; invoice DAS.
- **TACEDGE:** marketing routes and glass theme.

Inconsistent or unclear behaviour found

- Two incident stores (incidents table vs incident-report forms) with separate list surfaces.
- Three grouping units (projects / sites / jobs) with overlapping meaning.
- DAS gating by feature flag in one place and orgType in another.
- Settings page reachable by URL for non-admins (server still enforces; nav hides it).
- The public form-upload route is registered in both auth branches; its unauthenticated server path needs runtime confirmation.

26 SHEETS · EVERY ROUTE ACCOUNTED FOR

05 · Page Definition Sheets

One sheet per meaningful active screen. Small route variants share a sheet; the complete route list is in the appendix and in evidence/V1_Route_and_Screen_Inventory.csv.

SCREENSHOTS

No screenshots appear on these sheets. The application was not run during this inspection (see section 00), so no live captures exist and none were fabricated. Each sheet's structure is reconstructed from the page source, hooks and routes it names under Evidence.

`PDS-DSH` · Dashboard `IMPLEMENTED`

Route: `/`

Give each role a live summary of projects, forms, incidents and activity on login.

USER / ROLE

All roles (widgets vary by role group)

SURFACE

Responsive

PRIMARY USER QUESTION

"What is happening across my work right now?"

ENTRY POINTS

Default authenticated route; sidebar Home; mobile bottom-nav Home.

EXIT POINTS

Into projects, forms, incidents and meetings via widget links.

INFORMATION HIERARCHY

Primary: role-specific stat widgets. Secondary: active jobs / recent forms. Tertiary: incident alerts.

MAIN COMPONENTS

Role-gated widget set from rbac-config DASHBOARD_WIDGETS; LPC tenants render the bespoke LPCDashboard instead.

INPUTS

None (read-only).

ACTIONS

Navigate; refresh.

OUTPUTS

None.

STATES · VALIDATION · PERMISSIONS

Loading skeletons via React Query; management-overview endpoint restricted to management roles server-side. Empty/error states not exhaustively traced.

DATA READ

```
GET /api/dashboard/management-overview, /stats, /my-activity
```

DATA WRITTEN

```
—
```

DEPENDENCIES

dashboard.ts routes (SQL aggregate stats)

KNOWN LIMITATIONS

Two inline useQuery calls (extraction debt). LPC dashboard bypasses the design tokens.

RELATED CAPABILITIES

`V1-DSH-01` · `V1-DSH-02`

CONFIDENCE

HIGH — validation: Widget correctness per role unverified at runtime.

EVIDENCE · PDS-DSH

`PDS-PRJ` · Projects list `IMPLEMENTED`

Route: `/projects`

List, create and open projects (relabelled "Departments" for the port tenant).

USER / ROLE

Management / Supervisor / Viewer

SURFACE

Responsive

PRIMARY USER QUESTION

"Which projects exist and what state are they in?"

ENTRY POINTS

Sidebar; dashboard links.

EXIT POINTS

Project detail; create-project dialog.

INFORMATION HIERARCHY

Primary: project cards/list with status. Secondary: create action. Tertiary: SharePoint sync (feature-flagged).

MAIN COMPONENTS

useProjects hook; project cards; status badges.

INPUTS

New project details.

ACTIONS

Create project; open project; sync SharePoint (flag).

OUTPUTS

projects rows.

STATES · VALIDATION · PERMISSIONS

Create gated by canManageJobs (write). Port tenants see relabelled copy.

DATA READ

```
GET /api/projects
```

DATA WRITTEN

```
POST /api/projects
```

DEPENDENCIES

projects.ts routes

KNOWN LIMITATIONS

None noted.

RELATED CAPABILITIES

`V1-PRJ-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-PRJ

`PDS-PRJD` · Project detail workspace `IMPLEMENTED`

Route: `/projects/:id`

The project hub: tabs for anchors, designs/setup, face photos, map/surveys, documents, reports, DAS specs, meetings and team.

USER / ROLE

Management / Supervisor

SURFACE

Desktop-first, responsive

PRIMARY USER QUESTION

"What is the full state of this project?"

ENTRY POINTS

Projects list; dashboard; global search.

EXIT POINTS

Drill Summary, Report Builder, survey viewers, face-photo canvas.

INFORMATION HIERARCHY

Primary: tab strip + active tab content. Secondary: project header with status/code. Tertiary: setup dialogs.

MAIN COMPONENTS

useProjectNav; ProjectDesignTab (784 lines); ProjectReportsTab (842); FacePhotoAnchorMap; MapSurvey; job-brief components.

INPUTS

Setup entities (designs, mixes, methods, zones, sections); uploads.

ACTIONS

Configure setup; plan anchors (manual or AI import); manage team; upload documents/photos.

OUTPUTS

Setup rows; planned anchors; documents.

STATES · VALIDATION · PERMISSIONS

Write actions gated by canManageJobs; anchor-import restricted to OrgAdmin/PM roles.

DATA READ

```
GET /api/projects/:id, /config, face-photos, surveys, reports
```

DATA WRITTEN

```
Setup CRUD; planned anchors; imports
```

DEPENDENCIES

anchor-logs/config.ts; anchor-import.ts; job-brief.ts

KNOWN LIMITATIONS

Largest composite surface in the app; several child components exceed 500 lines.

RELATED CAPABILITIES

`V1-PRJ-02` · `V1-CFG-01` · `V1-CFG-02` · `V1-CFG-03` · `V1-PLN-01` · `V1-PLN-02` · `V1-MAP-02`

CONFIDENCE

HIGH

EVIDENCE · PDS-PRJD

`PDS-DRL` · Drill (field capture) `IMPLEMENTED`

Route: `/drill`

Capture drill and grout logs against planned anchors in a grid built for gloves-on field entry.

USER / ROLE

Field crew

SURFACE

Phone/tablet-first

PRIMARY USER QUESTION

"Which anchor am I on, and what do I need to record?"

ENTRY POINTS

Sidebar (Field group); mobile bottom-nav Drill button.

EXIT POINTS

Anchor detail dialogs; Drill Summary (management).

INFORMATION HIERARCHY

Primary: anchor grid by zone with status colours. Secondary: quick-fill dialogs. Tertiary: notes/issues.

MAIN COMPONENTS

useDrill hook; DrillerQuickFillDialog; GrouterQuickFillDialog; active-zone persistence (localStorage).

INPUTS

Drill method, depths, dates, lithology layers, grout mix/volumes, issues.

ACTIONS

Record drill; record grout; bulk drill/grout; flag issues.

OUTPUTS

anchor_install_logs updates; ground_profile_layers.

STATES · VALIDATION · PERMISSIONS

Offline drafts persist via IndexedDB/localStorage; sync queue replays on reconnect. Status auto-advances planned → drilled → grouted.

DATA READ

```
GET /api/projects/:id/anchor-logs, /config
```

DATA WRITTEN

```
POST/PATCH anchor-logs; bulk endpoints
```

DEPENDENCIES

offline-store.ts; anchor-logs routes

KNOWN LIMITATIONS

797 lines with inline logic; offline replay behaviour unverified at runtime.

RELATED CAPABILITIES

`V1-DRL-01` · `V1-DRL-02` · `V1-DRL-03` · `V1-GRT-01` · `V1-OFF-01`

CONFIDENCE

HIGH — validation: Field offline capture end-to-end.

EVIDENCE · PDS-DRL

`PDS-ALG` · Drill Summary (anchor logs review) `IMPLEMENTED`

Route: `/anchor-logs`

Review all anchor install logs across a project: filter, bulk-edit, flag for PM review, export XLSX and email the log set.

USER / ROLE

Management / Supervisor

SURFACE

Desktop-first

PRIMARY USER QUESTION

"Where is drilling up to, and is the record complete?"

ENTRY POINTS

Sidebar Project Operations.

EXIT POINTS

Anchor detail dialogs; XLSX download; email flow.

INFORMATION HIERARCHY

Primary: log table with status/test columns. Secondary: filters + bulk actions. Tertiary: export/email.

MAIN COMPONENTS

useAnchorLogsPage; AnchorLogsPdfDocument; archive list/restore.

INPUTS

Filters; bulk edits; recipient selection.

ACTIONS

Bulk update; archive/restore; export XLSX; email logs to a teammate.

OUTPUTS

XLSX file; email via Resend; log updates.

STATES · VALIDATION · PERMISSIONS

Writes gated by canManageForms; email recipients restricted to tenant teammates (zod-validated).

DATA READ

```
GET /api/projects/:id/anchor-logs; /api/org/teammates
```

DATA WRITTEN

```
PATCH anchor-logs; POST /api/anchor-logs/email
```

DEPENDENCIES

drill-log-xlsx.ts (Python-built template)

KNOWN LIMITATIONS

950 lines; 2 inline queries.

RELATED CAPABILITIES

`V1-DRL-03` · `V1-RPT-03`

CONFIDENCE

HIGH

EVIDENCE · PDS-ALG

`PDS-TST` · Anchor Testing `IMPLEMENTED`

Route: `/anchor-testing`

Record and review anchor load tests: readings per load step, photo OCR of paper sheets, AI conformance analysis, exports.

USER / ROLE

Management / Supervisor / Viewer

SURFACE

Responsive

PRIMARY USER QUESTION

"Did this anchor pass, against which standard?"

ENTRY POINTS

Sidebar (opt-in feature flag).

EXIT POINTS

Test XLSX / PDF exports.

INFORMATION HIERARCHY

Primary: test list with conformance. Secondary: test entry with dial readings. Tertiary: AI OCR/analysis panel.

MAIN COMPONENTS

useAnchorTestingPage; test photo upload.

INPUTS

Load steps, dial readings (1-3 gauges), standard, jack asset, photos.

ACTIONS

Create/edit test; OCR photo; run AI analysis; export.

OUTPUTS

anchor_test_results + readings; XLSX; engineering PDF.

STATES · VALIDATION · PERMISSIONS

Writes gated by canManageForms; AI endpoints require auth.

DATA READ

```
GET /api/anchor-tests; /api/anchor-logs/:logId/test-result
```

DATA WRITTEN

```
POST/PATCH test-result; test-photo-ocr
```

DEPENDENCIES

ai-test-analyzer.ts (Claude); assets (jack calibration)

KNOWN LIMITATIONS

Paper remains the primary capture medium; the platform digitises downstream.

RELATED CAPABILITIES

`V1-TST-01` · `V1-TST-02` · `V1-TST-03` · `V1-TST-04`

CONFIDENCE

HIGH — validation: OCR accuracy with real sheets.

EVIDENCE · PDS-TST

`PDS-DAS` · Daily Activity Sheets `IMPLEMENTED`

Route: `/daily-activity`

Capture per-shift crew and plant hours against a project, auto-totalled, exportable as PDF and invoice XLSX.

USER / ROLE

Field / Supervisor

SURFACE

Responsive

PRIMARY USER QUESTION

"Who and what worked today, for how long?"

ENTRY POINTS

Sidebar Project Operations (hidden for port).

EXIT POINTS

PDF/XLSX downloads.

INFORMATION HIERARCHY

Primary: sheet editor (crew rows, plant rows). Secondary: sheet list by date. Tertiary: export actions.

MAIN COMPONENTS

TimesheetDAS; DAS crew/plant pickers fed by das_* specs.

INPUTS

Crew hours, plant hours, shift metadata.

ACTIONS

Create/edit sheet; export PDF; export invoice XLSX.

OUTPUTS

forms rows (type=daily-activity); PDF; XLSX.

STATES · VALIDATION · PERMISSIONS

DAS specs gating differs (feature flag vs !isPortOnly) — two mechanisms.

DATA READ

```
GET /api/daily-activity-sheets; das-config/crew/plant
```

DATA WRITTEN

```
POST/PATCH sheets
```

DEPENDENCIES

das-specs.ts; pdfkit; xlsx

KNOWN LIMITATIONS

1111 lines — largest page in the codebase.

RELATED CAPABILITIES

`V1-DAS-01` · `V1-DAS-02` · `V1-DAS-03`

CONFIDENCE

HIGH — validation: Live invoice usage.

EVIDENCE · PDS-DAS

`PDS-SAF` · Safety report lists `IMPLEMENTED`

Routes: `/incidents` · `/near-misses` · `/site-inspections` · `/asset-assessments`

Four sibling list screens over form submissions of a given type, each launching the matching form template.

USER / ROLE

All safety roles

SURFACE

Responsive

PRIMARY USER QUESTION

"What has been reported, and what needs action?"

ENTRY POINTS

Sidebar Safety group.

EXIT POINTS

Fill-form for new reports; submission detail dialog.

INFORMATION HIERARCHY

Primary: submission list with status badges. Secondary: new-report action. Tertiary: filters.

MAIN COMPONENTS

Shared list pattern; SubmissionDetailDialog (812 lines).

INPUTS

New report via template.

ACTIONS

Create report; open detail; change status.

OUTPUTS

forms rows of the respective type.

STATES · VALIDATION · PERMISSIONS

Uses /api/form-templates/available to find the right template; inline apiRequest writes (extraction debt).

DATA READ

```
GET /api/incident-reports | /near-misses | /site-inspections | /asset-assessments
```

DATA WRITTEN

```
Form submissions
```

DEPENDENCIES

forms.ts routes

KNOWN LIMITATIONS

Four near-duplicate pages sharing an unextracted pattern.

RELATED CAPABILITIES

`V1-SAF-01` · `V1-SAF-02` · `V1-SAF-03`

CONFIDENCE

HIGH

EVIDENCE · PDS-SAF

`PDS-AST` · Asset Register `IMPLEMENTED`

Route: `/assets`

Maintain the register of jacks, gauges, vehicles, plant and safety equipment with certification/calibration expiry.

USER / ROLE

Management / Supervisor

SURFACE

Responsive

PRIMARY USER QUESTION

"Is this equipment certified and in service?"

ENTRY POINTS

Sidebar Safety group.

EXIT POINTS

Asset detail; expiring-assets view.

INFORMATION HIERARCHY

Primary: asset list with expiry indicators. Secondary: asset editor. Tertiary: attachments.

MAIN COMPONENTS

useAssets hook.

INPUTS

Asset details, cert dates, calibration values, attachments.

ACTIONS

CRUD assets; view expiring.

OUTPUTS

assets rows.

STATES · VALIDATION · PERMISSIONS

Authenticated; categories are free text.

DATA READ

```
GET /api/assets; /assets/expiring
```

DATA WRITTEN

```
POST/PATCH/DELETE assets
```

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

RELATED CAPABILITIES

`V1-AST-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-AST

`PDS-MTG` · Meetings list + recorder `IMPLEMENTED`

Route: `/meetings`

List safety meetings/toolbox talks and start recorded meetings with live transcription.

USER / ROLE

All roles

SURFACE

Responsive

PRIMARY USER QUESTION

"What meetings are planned or done, and can I start one now?"

ENTRY POINTS

Sidebar Safety group (SteepWorks re-points its sign-in item elsewhere).

EXIT POINTS

Meeting detail.

INFORMATION HIERARCHY

Primary: meeting list with status. Secondary: start/record action. Tertiary: stats.

MAIN COMPONENTS

MeetingRecorder (IndexedDB audio buffering); useMeetings.

INPUTS

Meeting metadata; microphone audio.

ACTIONS

Create meeting; record; upload audio.

OUTPUTS

safety_meetings rows; audio files; transcripts.

STATES · VALIDATION · PERMISSIONS

Recording disabled per-tenant via meetingVoiceRecording flag; capability canManageMeetings for writes.

DATA READ

```
GET /api/safety-meetings
```

DATA WRITTEN

```
POST meetings; POST /:id/audio
```

DEPENDENCIES

Deepgram; recording-idb.ts

KNOWN LIMITATIONS

644 lines.

RELATED CAPABILITIES

`V1-MTG-01` · `V1-MTG-02`

CONFIDENCE

HIGH

EVIDENCE · PDS-MTG

`PDS-MTGD` · Meeting detail `IMPLEMENTED`

Route: `/meetings/:id`

Run and review a meeting: attendance with signatures, tasks, photos, transcript and AI analysis, with live refresh.

USER / ROLE

All roles

SURFACE

Responsive

PRIMARY USER QUESTION

"Who attended, what was decided, and what actions came out?"

ENTRY POINTS

Meetings list.

EXIT POINTS

Export; incident links from tasks.

INFORMATION HIERARCHY

Primary: meeting record + transcript/analysis. Secondary: attendance list. Tertiary: tasks/photos.

MAIN COMPONENTS

useMeetingDetail; AttendanceList; MeetingDialogs; MeetingAnalysis.

INPUTS

Attendance signatures; tasks; questions to ask-AI.

ACTIONS

Sign attendance; add tasks; run AI analysis; ask AI; export.

OUTPUTS

Participants/tasks/photos rows; aiAnalysis jsonb.

STATES · VALIDATION · PERMISSIONS

Live view via refetchInterval; audio delete restricted to management roles.

DATA READ

```
GET /api/safety-meetings/:id + sub-resources
```

DATA WRITTEN

```
Participants/tasks/photos; analyze; transcript
```

DEPENDENCIES

meeting-ai-analyzer.ts (Claude); OpenAI Realtime session

KNOWN LIMITATIONS

577 lines.

RELATED CAPABILITIES

`V1-MTG-01` · `V1-MTG-03` · `V1-MTG-04`

CONFIDENCE

HIGH

EVIDENCE · PDS-MTGD

`PDS-FRM` · Forms gallery + fill form `IMPLEMENTED`

Routes: `/forms` · `/forms/:id`

Choose an available form template and complete it with dynamic fields, attachments and signatures.

USER / ROLE

Field crew / all roles

SURFACE

Phone-first

PRIMARY USER QUESTION

"Which form do I need, and what must I fill in?"

ENTRY POINTS

Sidebar Admin/Forms; safety list screens; SteepWorks daily sign-in nav item.

EXIT POINTS

Submissions; JSA sign-in child flows.

INFORMATION HIERARCHY

Primary: template gallery / dynamic field renderer. Secondary: attachments + signature. Tertiary: project/asset linkage.

MAIN COMPONENTS

dynamic-form-renderer; SignaturePad.

INPUTS

Field values per template schema; attachments; signatures.

ACTIONS

Submit form; save draft (offline).

OUTPUTS

forms rows; form attachments.

STATES · VALIDATION · PERMISSIONS

Drafts survive offline via IndexedDB form-drafts store; zod insert-schema validation server-side.

DATA READ

```
GET /api/form-templates/available; /api/forms/:id
```

DATA WRITTEN

```
POST /api/forms; attachments
```

DEPENDENCIES

form_templates.schema jsonb

KNOWN LIMITATIONS

formData shape varies by template (unstructured jsonb).

RELATED CAPABILITIES

`V1-FRM-01` · `V1-FRM-02` · `V1-SAF-04` · `V1-OFF-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-FRM

`PDS-FB` · Form Builder `IMPLEMENTED`

Route: `/form-builder`

Author custom form templates with drag-and-drop fields across 11 form types.

USER / ROLE

Management

SURFACE

Desktop-only

PRIMARY USER QUESTION

"What should this form ask, in what order?"

ENTRY POINTS

Sidebar Admin (default-on flag; flipped on for SteepWorks by migration 0046).

EXIT POINTS

Forms gallery.

INFORMATION HIERARCHY

Primary: field canvas. Secondary: field palette (7 types). Tertiary: template settings (daily sign-in toggle).

MAIN COMPONENTS

@dnd-kit sortable builder.

INPUTS

Template structure.

ACTIONS

Create/edit/delete templates.

OUTPUTS

form_templates rows.

STATES · VALIDATION · PERMISSIONS

Gated by canManageForms.

DATA READ

```
GET /api/form-templates
```

DATA WRITTEN

```
POST/PATCH/DELETE form-templates
```

DEPENDENCIES

—

KNOWN LIMITATIONS

574 lines.

RELATED CAPABILITIES

`V1-FRM-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-FB

`PDS-SUB` · Submissions inbox `IMPLEMENTED`

Route: `/submissions`

Review submitted forms across types with status lifecycle and archive filter.

USER / ROLE

All roles

SURFACE

Responsive

PRIMARY USER QUESTION

"What has been submitted and what state is it in?"

ENTRY POINTS

Sidebar Admin.

EXIT POINTS

Submission detail dialog; PDF render of a submission.

INFORMATION HIERARCHY

Primary: submission list. Secondary: detail dialog with attachments. Tertiary: archive filter.

MAIN COMPONENTS

SubmissionDetailDialog (812 lines); FormSubmissionPdf.

INPUTS

Status changes.

ACTIONS

Approve/reject/archive; export submission PDF.

OUTPUTS

forms status updates.

STATES · VALIDATION · PERMISSIONS

Status: draft → pending → approved/rejected → completed → archived.

DATA READ

```
GET /api/forms?includeArchived=true
```

DATA WRITTEN

```
PATCH /api/forms/:id
```

DEPENDENCIES

—

KNOWN LIMITATIONS

3 inline queries.

RELATED CAPABILITIES

`V1-FRM-03`

CONFIDENCE

HIGH

EVIDENCE · PDS-SUB

PDS-VAR · Variations IMPLEMENTED

Route: /variations

Raise contract variations with reason and status workflow; Teams notification on submission.

USER / ROLE

Management / Supervisor

SURFACE

Responsive

PRIMARY USER QUESTION

"What changed against the contract, and has the client approved it?"

ENTRY POINTS

Sidebar Project Operations (opt-in flag).

EXIT POINTS

Project variations list.

INFORMATION HIERARCHY

Primary: variation form. Secondary: status workflow. Tertiary: linkage to project.

MAIN COMPONENTS

VariationFormBuilder.

INPUTS

Variation details, reason.

ACTIONS

Draft; submit; approve/reject.

OUTPUTS

variations rows; Teams card.

STATES · VALIDATION · PERMISSIONS

draft → submitted → approved/rejected.

DATA READ

```
GET /api/projects/:id/variations
```

DATA WRITTEN

```
POST /api/variations
```

DEPENDENCIES

microsoftTeams.ts

KNOWN LIMITATIONS

None noted.

RELATED CAPABILITIES

V1-VAR-01

CONFIDENCE

HIGH

EVIDENCE · PDS-VAR

`PDS-RPT` · Project reports + Report Builder `IMPLEMENTED`

Routes: `/projects/:projectId/reports` · `/reports/:reportId`

Create engineer-facing reports from skeleton templates in a block editor, with AI-proposed sections, revision snapshots and signatories.

USER / ROLE

Management

SURFACE

Desktop-only (builder)

PRIMARY USER QUESTION

"Is the client report complete, reviewed and issuable?"

ENTRY POINTS

Project detail Reports tab; project reports list.

EXIT POINTS

PDF render; revision history.

INFORMATION HIERARCHY

Primary: Tiptap block editor. Secondary: data-bundle blocks (drill/test data). Tertiary: status + revision panel.

MAIN COMPONENTS

ReportEditor; blocks/data-blocks (760 lines); reports/theme.ts.

INPUTS

Report content; signatories.

ACTIONS

Edit; AI propose; snapshot revision; change status.

OUTPUTS

reports + report_revisions rows; PDF.

STATES · VALIDATION · PERMISSIONS

Text state machine enforced at route layer; revisions immutable.

DATA READ

```
GET /api/reports/:id, /data-bundle, /revisions
```

DATA WRITTEN

```
PATCH report; POST ai-propose; revisions
```

DEPENDENCIES

@tiptap; @react-pdf/renderer; ai-report-proposer.ts

KNOWN LIMITATIONS

Status not a DB enum; builder is desktop-oriented.

RELATED CAPABILITIES

`V1-RPT-01` · `V1-RPT-02`

CONFIDENCE

HIGH — validation: Whether reports are issued to clients in practice.

EVIDENCE · PDS-RPT

`PDS-DOC` · Documents (SharePoint browser) `IMPLEMENTED`

Route: `/documents`

Browse, search and download SharePoint drive content via Microsoft Graph.

USER / ROLE

Management / Viewer

SURFACE

Desktop-first

PRIMARY USER QUESTION

"Where is the project document I need?"

ENTRY POINTS

Sidebar (opt-in flag).

EXIT POINTS

File downloads.

INFORMATION HIERARCHY

Primary: drive/folder tree. Secondary: search. Tertiary: download actions.

MAIN COMPONENTS

useDocuments.

INPUTS

Search queries.

ACTIONS

Browse; search; download.

OUTPUTS

—.

STATES · VALIDATION · PERMISSIONS

Requires tenant Azure/SharePoint configuration; read capability canManageJobs.

DATA READ

```
GET /api/documents/drives/*
```

DATA WRITTEN

```
—
```

DEPENDENCIES

microsoft-graph.ts; SHAREPOINT_* env

KNOWN LIMITATIONS

Distinct from per-project project_documents store — two document surfaces.

RELATED CAPABILITIES

`V1-DOC-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-DOC

`PDS-CAL` · Calendar `IMPLEMENTED`

Route: `/calendar`

Manage events with optional Outlook sync (port tenant uses it for vessel/maintenance scheduling).

USER / ROLE

Management / Viewer

SURFACE

Responsive

PRIMARY USER QUESTION

"What is scheduled?"

ENTRY POINTS

Sidebar (opt-in flag).

EXIT POINTS

—.

INFORMATION HIERARCHY

Primary: calendar grid. Secondary: event editor. Tertiary: Outlook connect status.

MAIN COMPONENTS

useCalendar.

INPUTS

Events; OAuth consent.

ACTIONS

CRUD events; connect/disconnect Outlook.

OUTPUTS

calendar_events rows; Outlook events.

STATES · VALIDATION · PERMISSIONS

Event CRUD gated by canManageIntegrations; public OAuth callback route.

DATA READ

```
GET /api/calendar/status, events
```

DATA WRITTEN

```
POST/PATCH/DELETE events
```

DEPENDENCIES

MS Graph delegated OAuth

KNOWN LIMITATIONS

None noted.

RELATED CAPABILITIES

`V1-CAL-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-CAL

`PDS-NTS` · Notes `IMPLEMENTED`

Route: `/notes`

Keep project-scoped markdown notes with attachments, visibility and pinning.

USER / ROLE

Field / Supervisor

SURFACE

Responsive

PRIMARY USER QUESTION

"What did we observe that does not fit a form?"

ENTRY POINTS

Sidebar Project Operations.

EXIT POINTS

—.

INFORMATION HIERARCHY

Primary: note list/editor. Secondary: project filter. Tertiary: pin/visibility controls.

MAIN COMPONENTS

useNotes.

INPUTS

Markdown, attachments.

ACTIONS

CRUD notes; pin.

OUTPUTS

notes rows.

STATES · VALIDATION · PERMISSIONS

Authenticated.

DATA READ

```
GET /api/notes
```

DATA WRITTEN

```
POST/PATCH/DELETE notes
```

DEPENDENCIES

—

KNOWN LIMITATIONS

Attachments stored inline as data URLs in jsonb.

RELATED CAPABILITIES

`V1-NOT-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-NTS

`PDS-JOB` · Jobs list + job by code `PARTIAL`

Routes: `/jobs` · `/jobs/:code`

List H&S jobs and look one up by its short code (the pre-project grouping unit).

USER / ROLE

Field / Supervisor

SURFACE

Responsive

PRIMARY USER QUESTION

"Which job am I on?"

ENTRY POINTS

Sidebar.

EXIT POINTS

Job detail.

INFORMATION HIERARCHY

Primary: job list with progress/status. Secondary: code lookup.

MAIN COMPONENTS

useJobsList.

INPUTS

Job details (create gated).

ACTIONS

Open job; create (canManageJobs).

OUTPUTS

jobs rows.

STATES · VALIDATION · PERMISSIONS

Overlaps with projects; retained for H&S flows.

DATA READ

```
GET /api/jobs; /api/jobs/code/:code
```

DATA WRITTEN

```
POST/PATCH jobs
```

DEPENDENCIES

—

KNOWN LIMITATIONS

Semi-legacy grouping unit.

RELATED CAPABILITIES

`V1-LEG-02`

CONFIDENCE

HIGH — validation: Which tenants still run on jobs.

EVIDENCE · PDS-JOB

`PDS-SIT` · Sites list + site detail `PARTIAL`

Routes: `/sites` · `/sites/:id`

Manage sites with contacts and files from the earlier site-centric model.

USER / ROLE

Management / Supervisor

SURFACE

Responsive

PRIMARY USER QUESTION

"What sites do we operate, with whom?"

ENTRY POINTS

Sidebar.

EXIT POINTS

Site detail with map.

INFORMATION HIERARCHY

Primary: site list. Secondary: detail with contacts/files. Tertiary: map.

MAIN COMPONENTS

useSiteDetail.

INPUTS

Site details, contacts, files.

ACTIONS

CRUD (canManageJobs).

OUTPUTS

sites/site_contacts/site_files rows.

STATES · VALIDATION · PERMISSIONS

Semi-legacy; still fully routed.

DATA READ

```
GET /api/sites
```

DATA WRITTEN

```
POST/PATCH sites; contacts; files
```

DEPENDENCIES

—

KNOWN LIMITATIONS

Superseded in practice by projects.

RELATED CAPABILITIES

`V1-LEG-01`

CONFIDENCE

HIGH — validation: Which tenants still use sites.

EVIDENCE · PDS-SIT

`PDS-SET` · Settings `IMPLEMENTED`

Route: `/settings`

Administer the organisation: users, role-permission overrides, branding, integrations, DAS specs and org settings.

USER / ROLE

OrgAdmin (canManageUsers)

SURFACE

Desktop-first

PRIMARY USER QUESTION

"How is this organisation configured?"

ENTRY POINTS

Sidebar Admin (capability-gated); user dropdown.

EXIT POINTS

—.

INFORMATION HIERARCHY

Primary: settings tabs. Secondary: user table + invites. Tertiary: branding uploads / RBAC matrix editor.

MAIN COMPONENTS

BrandingTab; IntegrationsTab; role-permissions editor.

INPUTS

User accounts; capability overrides; branding assets; integration config.

ACTIONS

Manage users; edit RBAC overrides; upload branding; configure SharePoint.

OUTPUTS

users; role_capability_overrides; tenant_config; sharepoint_config.

STATES · VALIDATION · PERMISSIONS

Nav gated by canManageUsers; page itself not router-gated (server enforces).

DATA READ

```
GET /api/settings/*
```

DATA WRITTEN

```
PATCH settings endpoints
```

DEPENDENCIES

—

KNOWN LIMITATIONS

PATCH /api/tenant/config lacks zod validation; DAS specs tab hidden for port.

RELATED CAPABILITIES

`V1-USR-01` · `V1-USR-02` · `V1-ORG-02`

CONFIDENCE

HIGH

EVIDENCE · PDS-SET

`PDS-PRF` · Profile + Help `IMPLEMENTED`

Routes: `/profile` · `/help`

Edit own profile, install the PWA, and read tenant-aware help content.

USER / ROLE

All roles

SURFACE

Responsive

PRIMARY USER QUESTION

"How do I manage my account / get help?"

ENTRY POINTS

User dropdown in sidebar footer.

EXIT POINTS

—.

INFORMATION HIERARCHY

Primary: profile fields / help sections. Secondary: PWA install. Tertiary: support contacts (tenant).

MAIN COMPONENTS

useProfile; useTenant.

INPUTS

Profile fields.

ACTIONS

Update profile; install app.

OUTPUTS

users updates.

STATES · VALIDATION · PERMISSIONS

beforeinstallprompt suppressed globally; manual install from Profile.

DATA READ

```
GET /api/auth/user
```

DATA WRITTEN

```
PATCH /api/user/profile
```

DEPENDENCIES

—

KNOWN LIMITATIONS

None noted.

RELATED CAPABILITIES

`V1-OFF-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-PRF

`PDS-AUTH` · Authentication screens `IMPLEMENTED`

Routes: `/login` · `/register` · `/forgot-password` · `/reset-password` · `/request-access` · `(forced) change-password`

Sign in (password, Microsoft), self-register, recover access and rotate forced passwords, all tenant-branded pre-login.

USER / ROLE

Public / all users

SURFACE

Responsive

PRIMARY USER QUESTION

"How do I get in?"

ENTRY POINTS

Unauthenticated root (custom tenants); /login.

EXIT POINTS

Dashboard on success.

INFORMATION HIERARCHY

Primary: credential form. Secondary: OAuth button. Tertiary: recovery links.

MAIN COMPONENTS

Tenant branding via /api/tenant/branding-by-domain (public).

INPUTS

Credentials; emails.

ACTIONS

Login; register; reset; request access.

OUTPUTS

Sessions; users; access_requests.

STATES · VALIDATION · PERMISSIONS

Rate-limited 10/15 min; mustChangePassword forces the change screen.

DATA READ

```
branding-by-domain
```

DATA WRITTEN

```
auth endpoints
```

DEPENDENCIES

passwordAuth; microsoftAuth; Resend

KNOWN LIMITATIONS

LPC branding deliberately never returned by the public endpoint.

RELATED CAPABILITIES

`V1-AUTH-01` · `V1-AUTH-02` · `V1-AUTH-04` · `V1-AUTH-05`

CONFIDENCE

HIGH

EVIDENCE · PDS-AUTH

`PDS-UPL` · Public form upload `IMPLEMENTED`

Route: `/forms/:id/upload`

Let a device (possibly unauthenticated in the SPA) attach files to a specific form via a tokenised flow.

USER / ROLE

External / field

SURFACE

Phone-first

PRIMARY USER QUESTION

"How do I get this photo onto the form?"

ENTRY POINTS

Link/QR from a form.

EXIT POINTS

—.

INFORMATION HIERARCHY

Primary: upload dropzone. Secondary: form context.

MAIN COMPONENTS

useTenant.

INPUTS

Files.

ACTIONS

Upload.

OUTPUTS

Form attachments.

STATES · VALIDATION · PERMISSIONS

Registered in both authed and unauthed router branches; server-side attachment routes are authenticated — the reconciliation (FORM_UPLOAD_SECRET) needs runtime confirmation.

DATA READ

```
GET /api/forms/:id
```

DATA WRITTEN

```
POST /api/forms/:id/attachments
```

DEPENDENCIES

form-attachments.ts

KNOWN LIMITATIONS

Auth model for the unauthenticated path unclear from static reading.

RELATED CAPABILITIES

`V1-FRM-02`

CONFIDENCE

MEDIUM — validation: Confirm the unauthenticated upload flow end-to-end.

EVIDENCE · PDS-UPL

`PDS-MKT` · Marketing site (9 pages) IMPLEMENTED

Routes: `/ (landing)` · `/about` · `/geotech` · `/geospatial` · `/pricing` · `/case-studies` · `/contact` · `/experience` · `/try`

Present TACEDGE marketing content and capture contact/trial enquiries, inside the product SPA.

USER / ROLE

Public (TacEdge brand only)

SURFACE

Responsive

PRIMARY USER QUESTION

"What does TACEDGE offer?"

ENTRY POINTS

Public web on tacedgepm.com.

EXIT POINTS

Login / register / contact submission.

INFORMATION HIERARCHY

Primary: marketing layout with hero/features. Secondary: contact + trial forms.

MAIN COMPONENTS

MarketingLayout; MarketingNav; LandingHero.

INPUTS

Contact/trial details.

ACTIONS

Submit contact form.

OUTPUTS

Email via POST /api/contact.

STATES · VALIDATION · PERMISSIONS

Gated by brandName === "TacEdge"; custom tenants never see these.

DATA READ

```
—
```

DATA WRITTEN

```
POST /api/contact
```

DEPENDENCIES

contact.ts; Resend

KNOWN LIMITATIONS

Marketing bundled inside the product app (customer-specific).

RELATED CAPABILITIES

`V1-MKT-01`

CONFIDENCE

HIGH

EVIDENCE · PDS-MKT

THE V1 APPLICATION'S VISUAL SYSTEM — NOT THIS WEBSITE'S

06 · As-Built Design System

What the V1 application actually ships: tokens, type, components and status colouring, including where no coherent system exists. This is distinct from the visual system of this documentation site, which follows the V2 package styling.

HEADLINE FINDING

The repository's `design_guidelines.md` describes a **"Rock Control" red-and-slate Material Design system on Inter**. The shipped application is a **TACEDGE forest-green shadcn/ui ("new-york") system** with per-tenant CSS-variable theming and an Apple-style glass skin for the TACEDGE brand. The document is legacy and misleads new contributors. LEGACY

Design tokens (as built)

TOKEN	VALUE (LIGHT)	ROLE
`--primary`	`hsl(104 37% 20%)` — dark forest green	Brand primary; also --ring and sidebar primary
`--background` / `--card`	`210 14% 99%` / white	Page / raised surfaces
`--foreground`	`0 0% 23%`	Body text
`--border`	`0 0% 91%`	Hairlines
`--destructive` / `--warning` / `--success`	`0 70% 50%` / `38 92% 50%` / `142 71% 45%`	Signal colours
`--radius`	`0.5rem` (Tailwind lg 9px / md 6px / sm 3px; glass theme 0.75rem)	Corner radius
`--shadow-2xs … --shadow-2xl`	Layered HSL-alpha scale	Elevation; re-tuned per theme
tacedge palette (Tailwind)	forest `#2b4721` · olive `#6e7d5c` · sage `#b2b594` · cream `#fcf2e8`	Brand constants

Dark mode is class-based with a full parallel token set; the primary green is identical in both modes. Breakpoints are stock Tailwind (`sm/md/lg`); the documented desktop ≥1024 / tablet / mobile <768 intent maps onto them. The app shell locks to `h-dvh` with contained body scroll for mobile stability.

Logos and brand assets

- Per-tenant logo, favicon and primary colour come from `tenant_config`, applied at runtime by the useTenant provider (hex → HSL onto `--primary`, `--ring`, sidebar tokens) with document title and favicon swaps.
- Hostname-keyed pre-paint defaults are hard-coded for the four known brands so the login screen is on-brand before the API answers; the cache key has been version-bumped seven times (churn signal).

- Static assets ship in `client/public/` (per-tenant favicons: LPC SVG, SteepWorks PNG, TacEdge PNG).

Typography

FACE	WHERE DEFINED	ACTUAL USE
Inter	`--font-sans` on body	The working UI face across the application
Play	Tailwind `font-heading`	Marketing/landing surfaces primarily
Be Vietnam Pro	Tailwind `font-body`	Marketing/landing surfaces primarily
Menlo (mono)	`--font-mono`	Data readouts; `tabular-nums` styled in glass theme

Fonts load from Google Fonts via `client/index.html` (CSP allowlists fonts.googleapis.com / fonts.gstatic.com) — not self-hosted. No enforced type scale exists in config; the scale in design_guidelines.md is aspirational.

Component inventory

- **shadcn/ui:** 51 files in `components/ui/` (new-york style, neutral base) including custom additions `centred-dialog`, `combobox`, `signature-pad`.
- **Application components:** app-sidebar, top-bar, mobile-bottom-nav, search-command, dynamic-form-renderer, PullToRefresh, error-boundary; domain sets under `project/` (largest — FacePhotoCanvas 1,753 lines, plan designer, quick-fill dialogs, map/drone survey components), `meetings/`, `submissions/`, `dashboard/` (incl. LPCDashboard), `reports/` (block editor), `shared/map/`, `landing/`.
- **Icons:** lucide-react exclusively (212 files). **Toasts:** the shadcn Radix toast system, mounted once (88 usage sites); no sonner.

Status indicators

Status colouring is the least systematic part of the UI — three competing patterns coexist: `PARTIAL`

1. Badge `cva` variants (default/secondary/destructive/warning/success/outline) selected by at least four separately-duplicated `getStatusVariant()` functions.
2. Two near-identical form-status maps that have drifted (one omits `submitted`).
3. At least five independent raw-hex `STATUS_COLORS` maps for the anchor lifecycle (planned slate, drilled blue, grouted cyan, tested green, requires-redrill amber) used by canvas pins, plan designer, dashboards and PDF exports.

Inconsistencies and one-off styling

- `lib/dashboardTheme.ts` carries two palette intents at once: a "Rock Red" chart palette and a re-briefed "military/industrial matte charcoal" set, with dead legacy aliases kept so old callers compile.
- Dashboards (both generic KPI cards and LPCDashboard) hard-code hex gradients outside the token system.
- 117 inline `style=` usages across 50 files — largely legitimate canvas/SVG/PDF positioning, but a competing pattern.
- The glass theme (blur, gradients, float/pulse animations) activates only when `brandName === "TacEdge"`, so tenants experience structurally different design systems.

Accessibility evidence

Radix primitives provide baseline keyboard/ARIA behaviour, and shadcn patterns carry focus styles. No accessibility testing artefacts, audits or lint rules were found in the repository. UNVERIFIED

THE IMPLEMENTATION BENEATH THE SURFACES

07 · Technical & Data Architecture

Stack, structure, data, identity, integrations, operations and debt — separated into V1 observation, V1 constraint, and V2 scoping implication.

Architecture overview

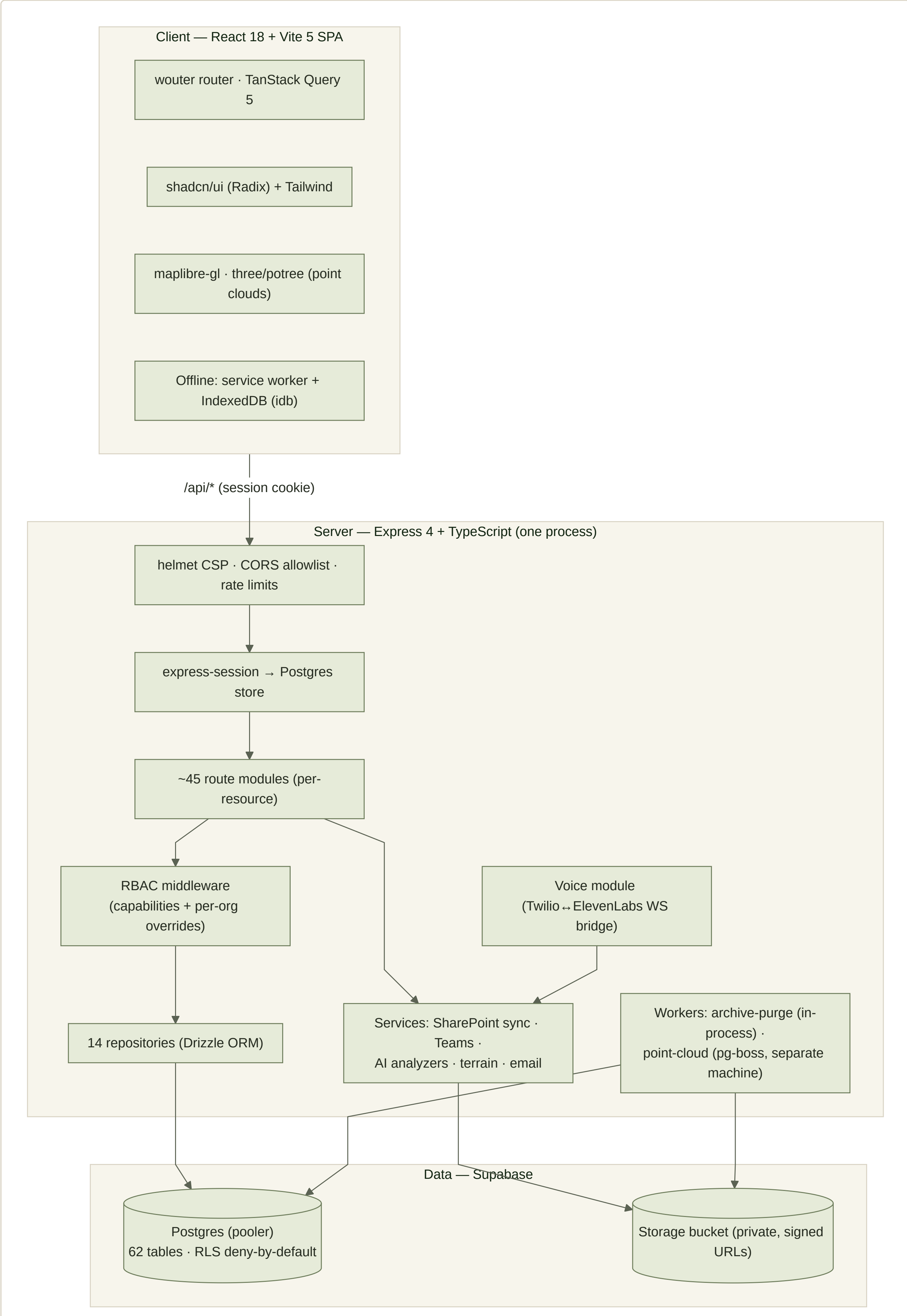
Client — React 18 + Vite 5 SPA
wouter router · TanStack Query 5
shadcn/ui (Radix) + Tailwind
maplibre-gl · three/potree (point clouds)
Offline: service worker + IndexedDB (idb)
/api/* (session cookie)
Server — Express 4 + TypeScript (one process)
helmet CSP · CORS allowlist · rate limits
express-session → Postgres store
~45 route modules (per-resource)
RBAC middleware (capabilities + per-org overrides)
Voice module (Twilio↔ElevenLabs WS bridge)
14 repositories (Drizzle ORM)
Services: SharePoint sync · Teams · AI analyzers · terrain · email
Workers: archive-purge (in-process) · point-cloud (pg-boss, separate machine)
Data — Supabase
Postgres (pooler) 62 tables · RLS deny-by-default
Storage bucket (private, signed URLs)

FIG-07-01 High-level architecture: a single Express process serves the API and the built SPA; Supabase provides Postgres and object storage. Source: diagrams/d07-system-architecture.mmd.

LAYER	TECHNOLOGY (VERSION)
Frontend	React 18.3 · Vite 5 · wouter 3.3 · TanStack Query 5.60 · Tailwind 3.4 + shadcn/ui (Radix) · maplibre-gl 5.18 · three 0.182 + potree-core 2.0 · @tiptap 3.23
Backend	Express 4 · TypeScript (ESM) · Drizzle ORM 0.39 · postgres 3.4 driver · zod 3.24
Data	Supabase Postgres (pooler) · Supabase Storage (one private bucket) · connect-pg-simple sessions
AI	@anthropic-ai/sdk 0.78 (claude-sonnet-4-6 · claude-haiku-4-5) · openai 6.9 (Realtime, one use) · Deepgram · ElevenLabs
Build/deploy	vite build + esbuild server bundle · Docker (node:20-slim + python3/openpyxl) · Fly.io syd · GitHub Actions CI

Application components and repository structure

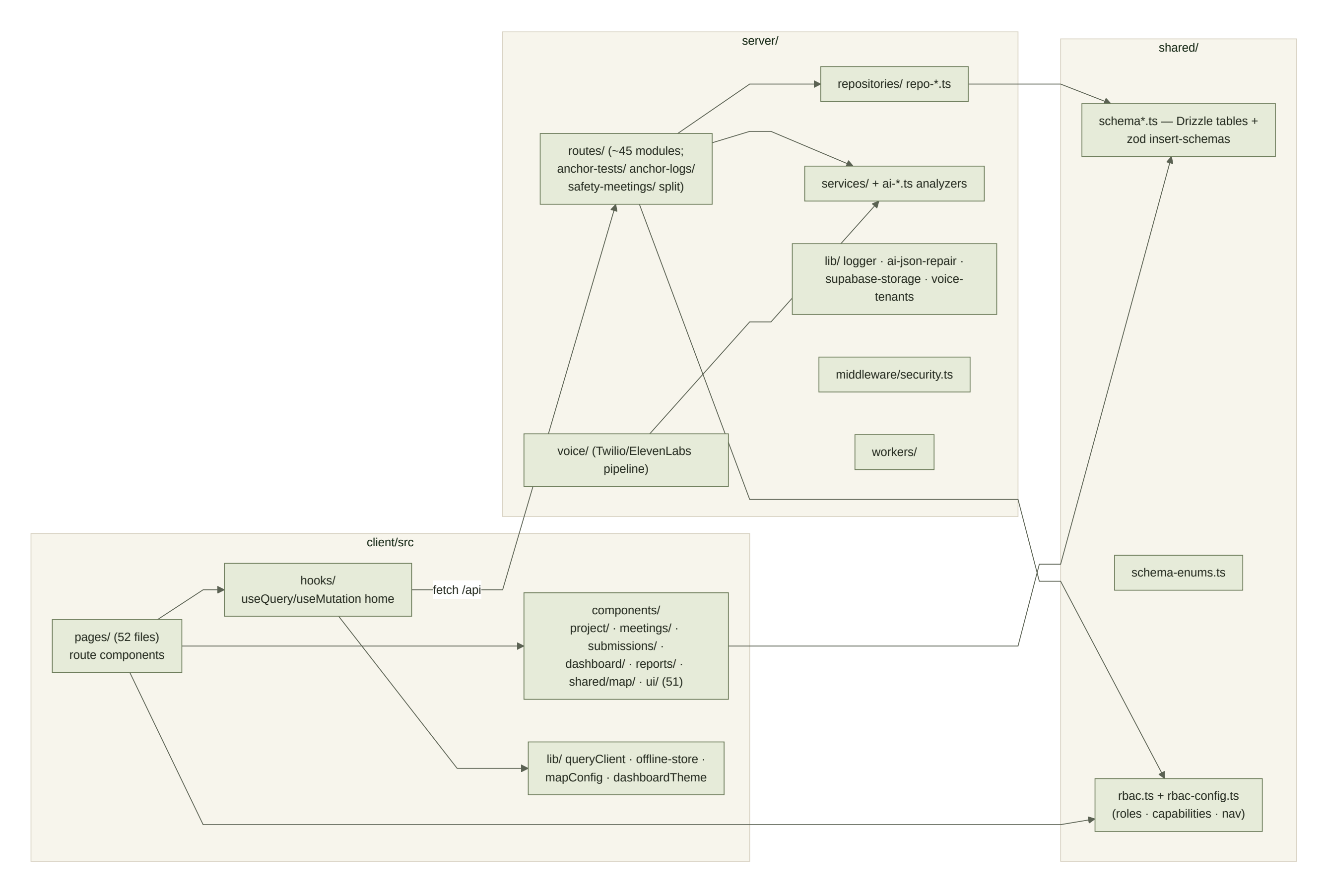

FIG-07-02 Component structure. shared/ carries schema, enums and RBAC config consumed by both sides. Source: diagrams/d08-app-components.mmd.

- **Routing:** ~45 route modules mounted in a single registry (server/routes.ts); large domains split into sub-module folders (anchor-tests/, anchor-logs/, safety-meetings/, reports/). All paths hard-code their `/api/...` prefixes.
- **State management:** TanStack Query with a 60 s global staleTime, IndexedDB persistence, and live views via refetchInterval. Client-local UI state only; no Redux-style store.
- **Storage layer:** 14 repository files composed into an IStorage facade; every query takes organizationId.

Data architecture

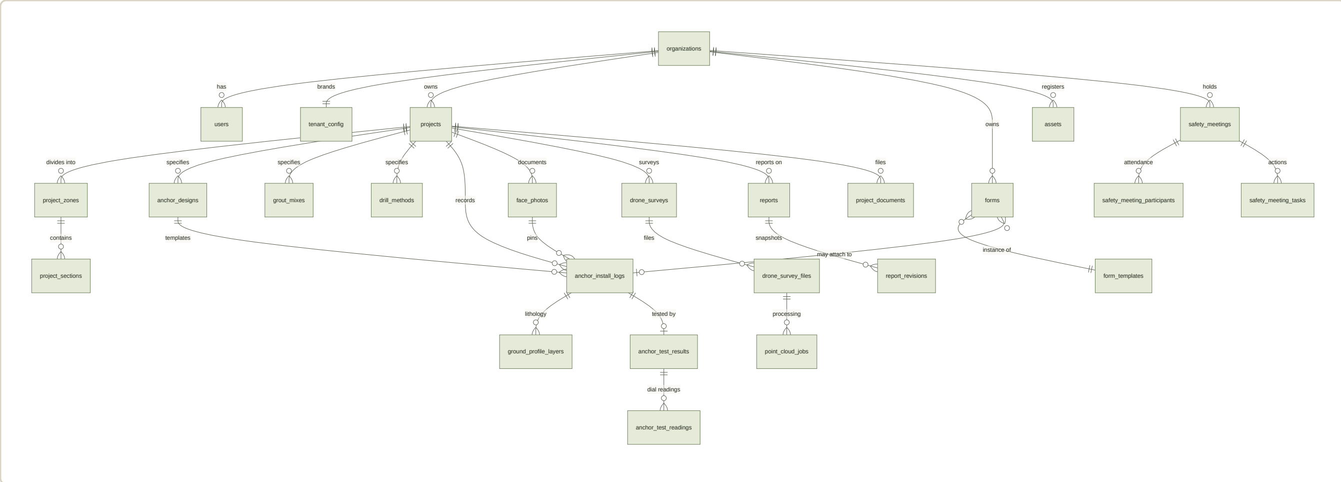

FIG-07-03 Core entity model (62 active tables; full inventory in the appendix and evidence CSV). Source: diagrams/d10-er-model.mmd.

- **Tenancy:** organization = tenant; no separate tenants table; orgType enum (engineering/port + dormant operator/council/mixed) shapes navigation.
- **Keys and audit:** UUID PKs; createdAt/updatedAt widespread; creator FKs; pervasive soft delete via deletedAt.
- **JSONB:** heavy use — form data/templates, AI responses, annotations, transcripts, feature flags, scope. Flexible but unvalidated at the DB layer.
- **Migrations:** hand-written SQL (idempotent by convention) tracked in schema_migrations and applied by the deploy release command; a parallel drizzle-kit push path created many tables with no CREATE migration file — the SQL history is incomplete by design.
- **RLS:** enabled on every public table, no policies (deny-by-default for Supabase API roles); the migrate script fails deploys that leave a table exposed. The Express role bypasses RLS — app-layer filtering is the real isolation.

Authentication, authorisation and permissions

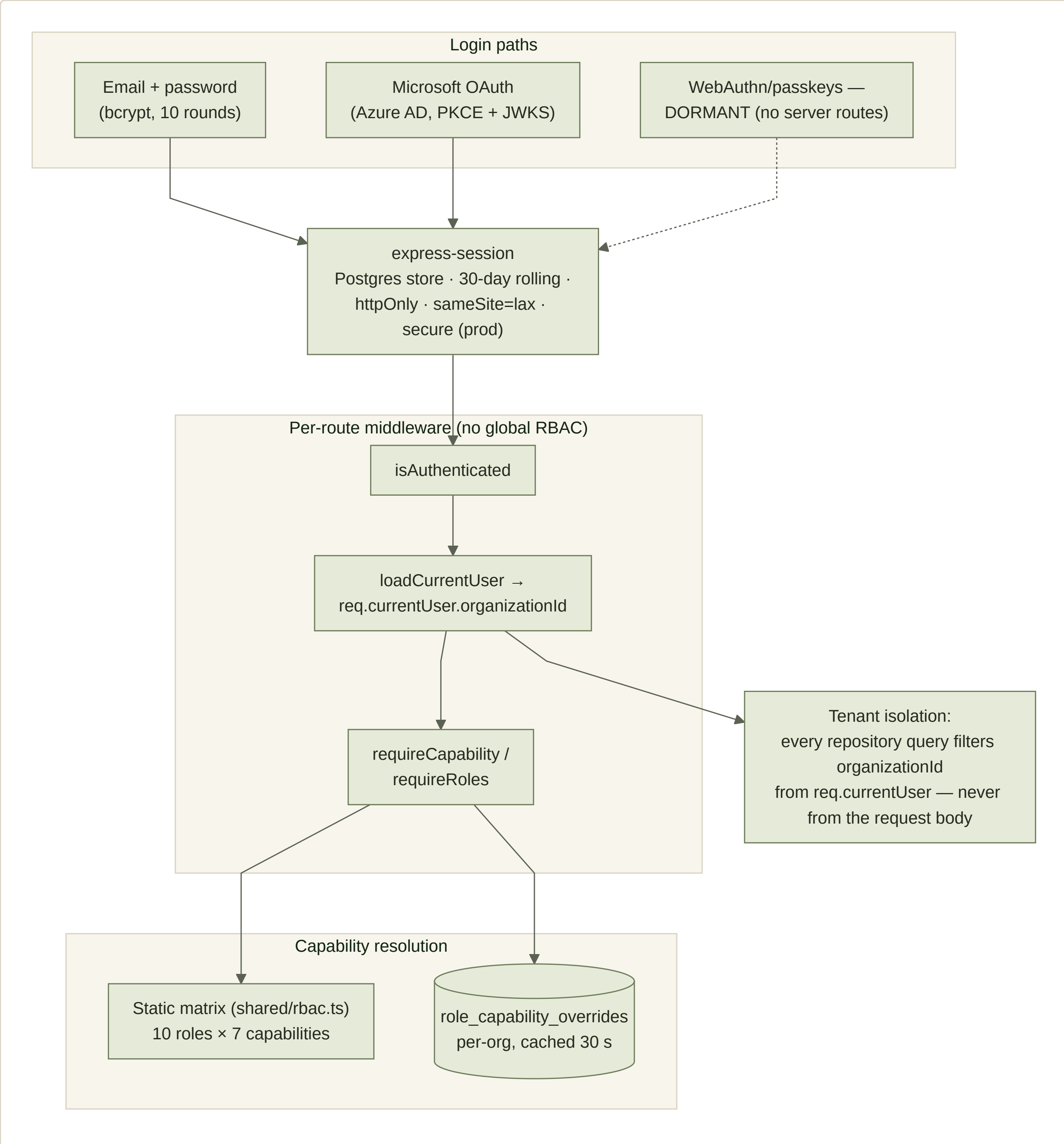

FIG-07-04 Identity and permission flow. WebAuthn is present in dependencies and schema but has no registered endpoints. Source: diagrams/d11-auth-permissions.mmd.

File and media storage

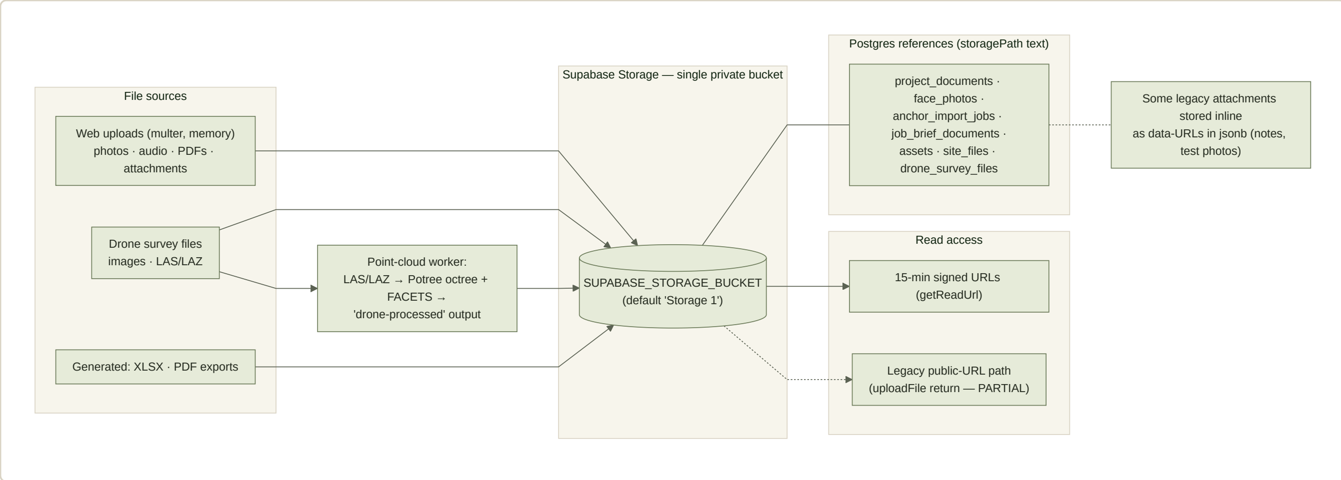

FIG-07-05 File architecture: one private bucket, storagePath references, 15-minute signed reads, plus a legacy public-URL path and inline data-URL attachments. Source: diagrams/d12-file-data-flow.mmd.

Voice architecture

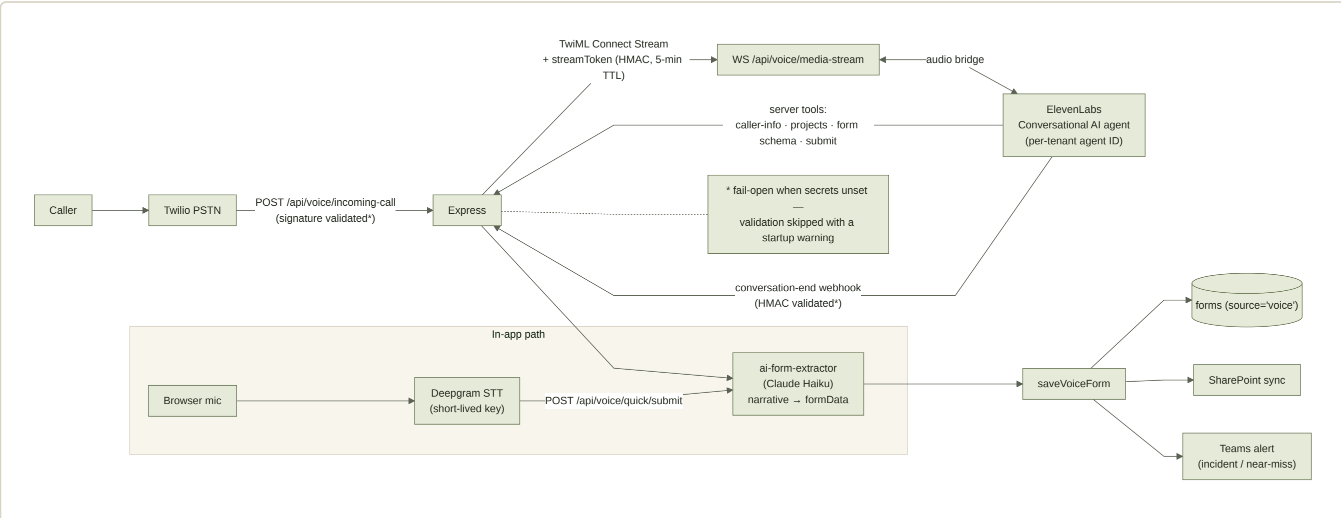

FIG-07-06 Voice intake. Phone and in-app paths converge on the same extraction and save pipeline. Source: diagrams/d13-voice-pipeline.mmd.

Deployment and environments

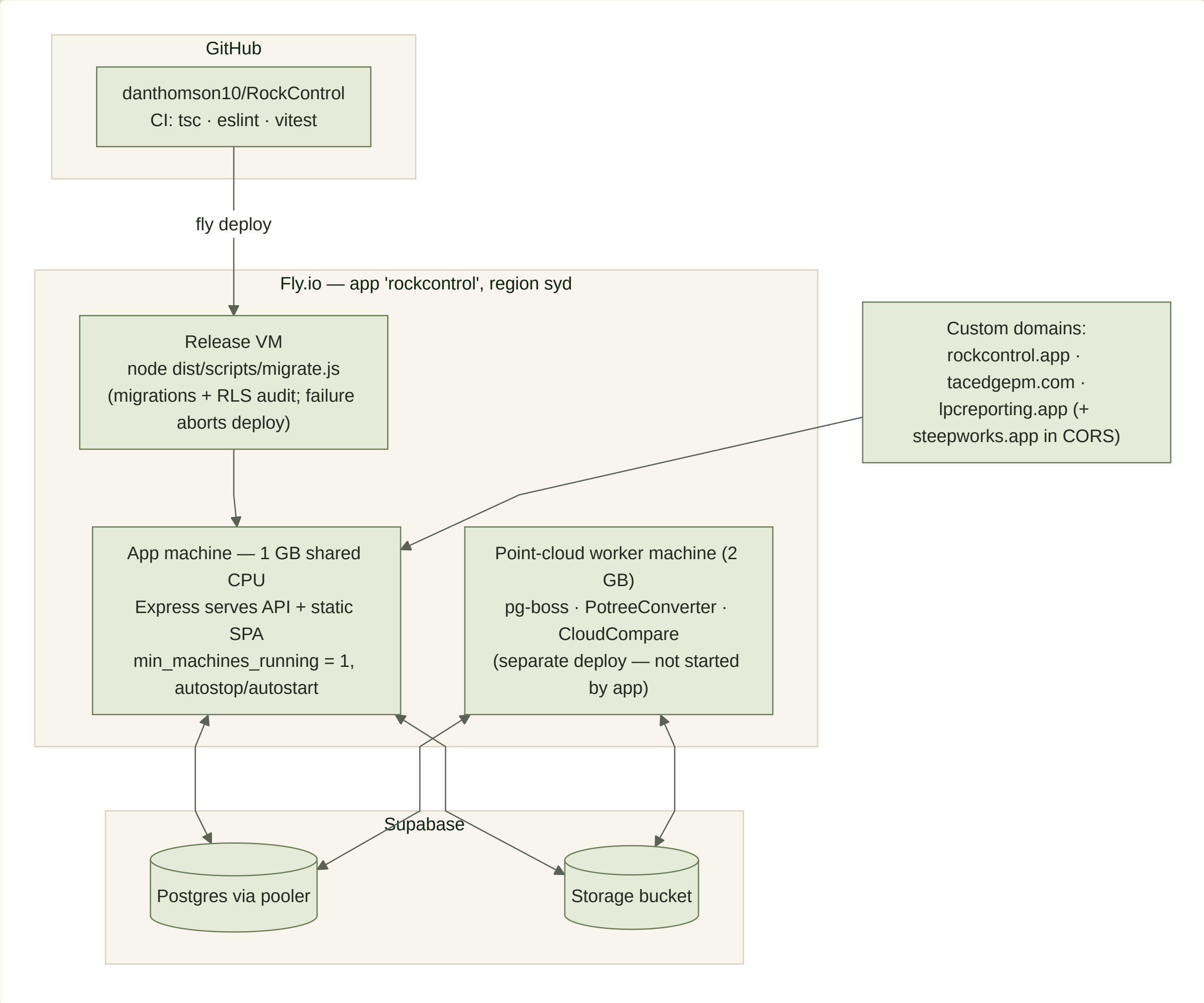

FIG-07-07 Deployment: one 1 GB shared-CPU machine (min 1 running), migration release-command gate, optional separate point-cloud worker machine. Source: diagrams/d09-deployment.mmd.

- **Environments:** production only, as far as the repository shows. No staging config exists; local dev needs a personal Supabase and .env. UNVERIFIED whether any shared non-prod environment exists outside the repo.
- **Configuration:** ~50 env vars (names in the appendix). Only DATABASE_URL and SESSION_SECRET fail closed; a written zod validator (server/lib/config.ts) is never invoked. Webhook/Twilio validation fails open when unset.
- **Server timeout** raised to 5 minutes for long AI requests — significant on a single shared VM.

Error handling, logging, monitoring, testing

- Central async-handler + errMessage pattern; structured console logger (LOG_LEVEL) to Fly logs.
- **No Sentry/APM/metrics/analytics.** docs/observability.md marks them future work. NOT IMPLEMENTED
- Tests: 8 vitest files (AI modules, JSON repair, one route) run in CI with tsc + eslint --max-warnings 0; Playwright smoke spec exists but @playwright/test is undeclared, so e2e is not runnable. PARTIAL
- Backups: no evidence beyond Supabase defaults. UNVERIFIED

Security and privacy posture

- Strong: strict production CSP (no unsafe-inline scripts), CORS allowlist with credential rules, rate limiting by endpoint class, hashed reset tokens, HMAC-bound voice stream tokens (5-min TTL), AI prompt sanitisation (sanitizeForPrompt/safeBlock), RLS + deploy audit.
- Weaker: fail-open webhook/Twilio validators; un-zodded PATCH /api/tenant/config; @rockcontrol.app verification bypass; committed one-off scripts naming customer admins; attached_assets/ needing a PII/security pass; single Teams webhook URL for all tenants.
- Privacy: personal data (names, emails, signatures, voice transcripts, phone callers) is stored per-tenant; no retention policy is visible beyond the 30-day purge of archived anchor logs.

Technical debt, fragile and coupled areas

AREA	OBSERVATION	V2 SCOPING IMPLICATION
Oversized modules	FacePhotoCanvas 1,753 · repo-features 1,197 · daily-activity 1,111 · FacePhotoAnchorMap 1,088 · anchor-logs 950 (20 files > 500 lines); several regrew after documented refactors	Cost of change is highest exactly where V2's field UX would differ most
Hook-extraction rule collapsed	124 inline useQuery/useMutation sites across 50 files, suppressed by eslint-disable TODOs	Client data layer is diffuse; hard to lift wholesale
Tenant branching in code	brandName/orgType string checks in App, nav, dashboards, email, voice, security middleware	"Config-not-code" needs a real boundary in V2
Status vocabularies duplicated	≥5 hex maps + ≥4 variant functions + 2 drifted form maps	A single status model is a V2 design decision, not a refactor
Stale internal docs	CLAUDE.md sizes/claims outdated; design_guidelines.md defunct; WEBHOOK_CONFIG.md stale	Trust code over docs during scoping
Dual grouping + incident models	projects/sites/jobs; incidents table vs forms	Data migration scope for V2 is ambiguous until resolved

Reusable parts of V1

- **High reuse potential:** the schema's geotech core (projects/zones/designs/logs/tests with enums), the repository-layer tenant-isolation pattern, the RLS + migration-audit machinery, AI prompt-sanitisation and JSON-repair library, the voice pipeline design, XLSX/PDF export logic, and the offline sync-queue design.
- **Reusable with caution:** RBAC config model (carries aviation residue), form-template system (unvalidated jsonb), shadcn component layer (status colouring fragmented).
- **Likely replace or rework:** oversized page/canvas components, dashboards, tenant branching, sites/jobs models, marketing-in-SPA, observability (absent), e2e harness.

Decisions that cannot be established from the repository

- Why sites/jobs were kept alongside projects; why incidents live twice.
- Whether the point-cloud worker ever ran in production.
- Original intent of the aviation vertical and Co-Pilot, and why both were dropped.
- Capacity headroom, cost profile, and real reliability of the single-VM deployment.

Decisions the V2 scoping sprint must make

1. Platform continuity: extend this Express/Supabase/Fly stack, or re-platform — and either way, what carries over (schema core? repositories? voice?).

2. The review lifecycle: how V2's submit/confirm/re-open model maps onto (or replaces) status + needsPmReview + form statuses.
3. Work-type configurability: whether the anchor-specific schema generalises or a work-item abstraction replaces it.
4. Tenancy: one multi-vertical codebase vs a ground-engineering workspace; what happens to LPC/port.
5. Data migration: which of the 62 tables' data must survive into V2.
6. Operational baseline: observability, staging, backup verification — currently absent, and priced into any option.

A NAVIGATION AID — NOT THE DELTA ASSESSMENT

08 · V1 → V2 Orientation

For each major product area: what V1 has (from this package's evidence), what the V2 reference proposes (from the V2 Product Definition PDF), the primary nature of the difference, and the questions the Technical Lead must answer. This section does not decide extend vs refactor vs rebuild.

HOW TO READ THIS

V1 claims cite this package's evidence base (TACEDGE Geotech @ 975a2ea). V2 claims come only from TACEDGE_Ground_Engineering_V2.0_Product_Definition.pdf. Neither is evidence of the other.

AREA	V1 (AS BUILT)	V2 (AS PROPOSED)	DIFFERENCE IS PRIMARILY	TECHNICAL LEAD MUST ANSWER
Product boundary	One multi-vertical, multi-tenant app: geotech + H&S + port skin + marketing site.	One Operational Workspace (B1) for ground engineering on a TACEDGE platform; six-work-type catalogue.	Product scope · platform configuration	Does V2 carve the workspace out of V1, or start beside it? What happens to port/H&S tenants?
Work types	Anchors hard-coded end-to-end; piling/drilling/shotcrete etc. absent (six nav placeholders unimplemented).	Work type = completion-criteria configuration on one spine; anchoring + piling proven in V2.0; four more onboard as config.	Data model · architecture	Can anchor_install_logs generalise to a work-item abstraction, or is that a new core?
Status & review model	Single anchor status enum (planned → drilled → grouted → tested + off-path) + needsPmReview flag; form statuses separate; no review lifecycle.	Two lifecycle dimensions (work phase + review state: in-progress → submitted → confirmed → re-opened) with testing and freshness overlays; single-owner editability per state; confirmed-only-outward.	Workflow · data model	Is the review dimension additive to V1's schema, or does it force re-modelling of logs and forms?
Engineer (client) surface	ClientViewer role with read capabilities; no designed engineer view; exports are emailed files.	A glance-first engineer view of confirmed records only; QA queue feeding it; closeout package from the live record.	Product scope · user experience	What must exist server-side (state, permissions, release gate) before an engineer view is honest?
PM status views	Drill Summary table + dashboards; face-photo anchor map (pins on photos); MapLibre project map.	Status Board (schematic by zone) + Spatial Map (work items on an uploaded site image); three layers: inform / act / release.	User experience	Do the face-photo canvas and zone model satisfy the Status Board/Spatial Map jobs, or is that surface new?

AREA	V1 (AS BUILT)	V2 (AS PROPOSED)	DIFFERENCE IS PRIMARILY	TECHNICAL LEAD MUST ANSWER
Safety / sign-on	JSA sign-ins recorded against parent forms; nothing gates drilling; full H&S suite (meetings, incidents, inspections) beyond V2's scope.	Daily JSA/SWMS sign-on with finger signature gates drill-log start (the one enforced barrier); incident/near-miss in <60 s.	Workflow	Where does the gate live (client, API, both) and does V1's form model support it? What of V1's wider H&S suite does V2 keep?
Offline	Service worker + IndexedDB sync queue with idempotency and retry; depth of coverage untested; runtime unverified.	DoD-level: a full shift with no signal and nothing lost on resync; offline-available project reference documents.	Reliability	Test V1's queue against the full-shift standard; decide if it is a foundation or a liability.
AI assistance	Broad: plan import, test OCR + conformance, incident/meeting/drone analysis, report proposals, voice extraction, map chat, Ask RC.	Narrow and disciplined: PDF plan import and test-record conversion, human-approved; voice transcription a Should.	Product scope	Which V1 AI features earn a place in V2's propose-approve-commit discipline; which are demos to retire?
Documents	Two surfaces: per-project uploads (Supabase) and a SharePoint browser (Graph).	A bounded project reference folder, offline-available, deliberately not a DMS; JSA/SWMS kept separate under sign-on.	Product scope · user experience	Consolidate to one document model? Does SharePoint remain a customer requirement?
Daily records	DAS with org rate cards (positions/personnel/plant), PDF + invoice XLSX.	Resourcing sheet: crew and plant hours, auto-totalled, exported; pay/invoice computation is roadmap.	Product scope	V1 already exceeds V2 here — does V2 adopt DAS specs as-is?
Voice	Full phone agent (Twilio/ElevenLabs) + in-app dictation, in production for incident-type forms.	Voice transcription for notes is a V2.0 Should; no phone agent in scope.	Product scope	Is the phone agent a retained V1 service, a V2 differentiator, or out of the workspace entirely?
Reporting	XLSX/PDF exports + Tiptap report builder with revisions and AI proposals; emailed delivery.	Reporting as release: confirmed records reach the engineer; closeout assembled from the live record.	Workflow · reporting	Does the report builder become the closeout tool, or does release-from-record replace document assembly?
Platform & ops	Express/Supabase/Fly monolith, one VM, no observability, production-only, SQL migrations with RLS audit.	Technical & Data Architecture artefact defines data/identity/state/interfaces; Operational Data Contract seam; coordination-readiness kept reachable.	Architecture · reliability	Stack continuity; where the Operational Data Contract would sit relative to V1's schema; minimum ops baseline for V2.

VALIDATION REQUIRED — INVESTIGATED FIRST, ASKED SECOND

09 · Open Questions

Only questions that materially affect understanding of V1 or the V2 scoping decision, raised after inspection could not resolve them. The same list ships as evidence/V1_Open_Questions_and_Validation_Required.md. Directed to Mike / Dan unless noted.

Product intent

1. What is the intended relationship between projects, sites and jobs? Three grouping units coexist; is the long-term model project-centric with sites/jobs retired?
2. Which incident store is authoritative — the dedicated incidents table or form-based incident reports? Both are routed.
3. Was the aviation vertical abandoned permanently, or might it return? It was fully built (migrations 0005–0012) then dropped (0055) within months; roles, org types and nav config remain.
4. What was Co-Pilot ("the brain") meant to become? Trigger-based event streaming was built and dropped within weeks.

Customer use

5. Which tenants are commercially active today, and at what volume? The repository cannot show usage.
6. Is the LPC engagement ongoing? Substantial port-specific behaviour is carried in the core codebase.
7. What is the status of the SteepWorks pilot ("pilot starts 25 May 2026" in a CSP comment)? Did it convert?
8. Is Groundline Civil (the V2 beachhead partner) a V1 user? No Groundline configuration was found in V1.

User roles

9. Are ClientViewer accounts actually issued to engineers/clients? The role exists; no designed engineer surface does.
10. Who uses the Subcontractor role, and how does it differ from FieldTech in practice? The capability matrix treats them identically.

Production behaviour

11. Has offline capture been verified end-to-end in the field? The sync queue exists; no test covers replay; runtime is UNVERIFIED here.
12. Is the point-cloud worker machine deployed? Nothing in the repo confirms one exists.
13. What are real voice-call completion rates on the Twilio/ElevenLabs pipeline?
14. How often do AI extractions (voice forms, test-sheet OCR, plan import) need correction? Accuracy decides whether these are capabilities or assisted drafts.

Reporting

15. Are Report Builder reports actually issued to clients? The revision/signatory model implies formal issue.

16. Which export (XLSX drill log, test PDF, DAS invoice) is the commercially load-bearing deliverable per tenant?

Data

17. How much production data sits in the semi-legacy tables (sites, jobs, take-5-era forms)? This sizes any V2 migration.
18. Are inline data-URL attachments (notes, test photos) materially large? They bypass the storage model.
19. What backup/recovery arrangements exist beyond Supabase defaults? None are evidenced in the repository.

Security

20. Are all webhook secrets and TWILIO_AUTH_TOKEN set in production? Validators fail open when unset.
21. Has attached_assets/ (107 files of pasted logs and customer documents) been reviewed for sensitive content?
22. What is the intent of the @rockcontrol.app email-verification bypass (server/routes/auth.ts:146)?
23. Is the single Supabase bucket confirmed private, and are legacy public-URL callers gone?

Infrastructure

24. Is one shared 1 GB VM adequate for current load, given 5-minute AI request timeouts? No metrics exist.
25. What is the environment story? No staging is evident — how are changes verified before production?
26. Where do secrets live and how are they rotated? Confirm docs/secret-rotation.md reflects practice.

Technical debt

27. Are the 500-line and hook-extraction rules still policy? Violations have regrown past documented refactors with lint suppressions.
28. Should the e2e harness be revived? Playwright config exists but the dependency was never declared.

Customer-specific logic

29. Which port/LPC behaviours are product capabilities vs one-off customisation for V2 scoping purposes?
30. Should the TACEDGE marketing site remain inside the product SPA?

V1-to-V2 scoping implications

31. Which V1 capabilities are contractually load-bearing for existing customers during any transition (voice intake, DAS invoicing, SharePoint sync)?
32. Is the V2 review model (submitted → confirmed → engineer visibility) intended to replace V1's status + needsPmReview model? V1 has no confirmed-only-outward gate today.
33. Does V2 inherit the Supabase/Fly/Express stack, or is the platform choice open? This determines V1 reuse.
34. Which tenants must migrate to V2, and which can remain on V1? Multi-tenant coupling is the largest structural difference from the V2 single-workspace definition.

INVENTORIES · METHODOLOGY · GLOSSARY

10 · Appendices

Compact renderings of the full inventories. The complete machine-readable versions live in the tacedge-v1-PD repository under `evidence/`.

A. Route inventory

ROUTE(S)	SCREEN	ROLE	CLASSIFICATION
`/`	Landing (TacEdge) or Login (custom tenant); Dashboard when authenticated	Public / all	IMPLEMENTED
`/login · /register · /forgot-password · /reset-password · /request-access`	Authentication screens	Public	IMPLEMENTED
`/about · /geotech · /geospatial · /pricing · /case-studies · /contact · /experience · /try`	Marketing pages (TacEdge brand only)	Public	IMPLEMENTED
`/forms/:id/upload`	Form upload (dual-registered public + authed)	External / field	IMPLEMENTED
`/projects · /projects/:id`	Projects list · project workspace	Mgmt / Supervisor	IMPLEMENTED
`/jobs · /jobs/:code`	Jobs list · job by code	Field / Supervisor	PARTIAL (semi-legacy)
`/sites · /sites/:id`	Sites list · site detail	Mgmt / Supervisor	PARTIAL (semi-legacy)
`/drill`	Field drill & grout capture	Field	IMPLEMENTED
`/anchor-logs`	Drill Summary review + exports	Mgmt / Supervisor	IMPLEMENTED
`/anchor-testing`	Anchor load testing (flag)	Mgmt / Supervisor / Viewer	IMPLEMENTED
`/daily-activity`	Daily activity sheets	Field / Supervisor	IMPLEMENTED
`/notes`	Project notes	Field / Supervisor	IMPLEMENTED
`/incidents · /near-misses · /site-inspections · /asset-assessments`	Safety report lists (4)	Safety roles	IMPLEMENTED
`/assets`	Asset register	Mgmt / Supervisor	IMPLEMENTED
`/meetings · /meetings/:id`	Safety meetings list · detail	All	IMPLEMENTED
`/documents`	SharePoint document browser (flag)	Mgmt / Viewer	IMPLEMENTED

ROUTE(S)	SCREEN	ROLE	CLASSIFICATION
/calendar	Calendar / Outlook sync (flag)	Mgmt / Viewer	IMPLEMENTED
/projects/:projectId/reports · /reports/:reportId	Project reports · Report Builder	Mgmt	IMPLEMENTED
/forms · /forms/:id	Forms gallery · dynamic fill	All / Field	IMPLEMENTED
/form-builder	Form template builder	Mgmt	IMPLEMENTED
/variations	Contract variations (flag)	Mgmt / Supervisor	IMPLEMENTED
/submissions	Submissions inbox	All	IMPLEMENTED
/settings	Organisation settings	OrgAdmin	IMPLEMENTED
/profile · /help	Profile · Help	All	IMPLEMENTED
(forced) change-password	Forced password rotation	All	IMPLEMENTED
* (authed)	Not Found	All	IMPLEMENTED
(none) admin-users.tsx	User admin table — never routed	Mgmt	DORMANT
(none) take-5-form.tsx	Hard-coded Take-5 — superseded	Field	LEGACY
/soil-nail-logs … /lab-testing (6)	Nav placeholders, no components	–	NOT IMPLEMENTED
/aviation/* (10 nav entries)	Dropped aviation module	–	LEGACY

Full detail: `evidence/V1_Route_and_Screen_Inventory.csv`.

B. Database objects by domain

DOMAIN	TABLES	SIGNAL
Tenancy & identity	organizations · tenant_config · users · sessions · clients · password_resets · email_verifications · access_requests · oauth_connections · webauthn_credentials · notification_preferences · role_capability_overrides	Active (webauthn_credentials dormant)
Projects & setup	projects · project_zones · project_sections · project_team_members · project_documents · anchor_designs · grout_mix_types · grout_mixes · drill_methods · drillers · signatories · caller_project_mappings	Active
Anchor record	anchor_install_logs · ground_profile_layers · anchor_test_results · anchor_test_readings · face_photos · anchor_import_jobs · job_briefs · job_brief_documents · variations	Active — the geotech core
DAS (rates & rosters)	das_positions · das_home_bases · das_personnel · das_plant · das_project_crew · das_project_plant · das_project_config	Active (added 0054, June 2026)
Forms & safety	forms · form_templates · incidents · assets	Active (dual incident stores)
Meetings	safety_meetings · safety_meeting_participants · safety_meeting_tasks · safety_meeting_photos	Active
Sites & jobs (older model)	sites · site_contacts · site_files · jobs · job_members	Partial / semi-legacy
Drone & spatial	drone_surveys · drone_survey_files · point_cloud_jobs · map_annotations · calendar_events	Active (point_cloud_jobs partial)
Reporting	reports · report_revisions	Active

DOMAIN	TABLES	SIGNAL
Integrations	sharepoint_config	Active
Dropped (0055)	16 aviation_* + 4 council_* + co_pilot_events + co_pilot_settings (+ 7 enums, 9 triggers, 2 functions)	Legacy — removed from DB

62 active tables, 30+ enums, one storage bucket, RLS everywhere. Full per-object detail including key fields, relationships and confidence: `evidence/V1_Database_Object_Inventory.csv`.

C. Key data dictionary (core record)

FIELD (ANCHOR_INSTALL_LOGS)	MEANING
anchorReference	Human anchor ID (e.g. A-12) within a zone/section
status	anchor_status enum: planned · drilled · grouted · tested · not-required · requires-redrill
anchorDesignId	FK to the design template supplying specs
testType	none · proof · performance · acceptance · investigation · suitability
issues / attachments	jsonb arrays: declared exceptions; file references
facePhotoId / redrilledFrom	Spatial pin on a face photo; link to the anchor this one replaces
needsPmReview	Review flag (boolean) — V1's only review construct
deletedAt	Soft delete; archived rows purge after 30 days

D. External dependencies

54 notable runtime dependencies grouped and version-listed in `evidence/V1_Dependency_Inventory.csv`. Highlights: React 18 / Vite 5 / wouter / TanStack Query 5 / Radix + Tailwind; maplibre + three/potree; Drizzle 0.39 on postgres 3.4; @anthropic-ai/sdk; twilio + ws + resend; @react-pdf + pdfkit + xlsx; pg-boss (partial); @simplewebauthn (dormant); @playwright/test (missing declaration).

E. Environment variables (names only)

GROUP	NAMES
Database / storage	DATABASE_URL · SUPABASE_URL · SUPABASE_SERVICE_ROLE_KEY · SUPABASE_STORAGE_BUCKET · SUPABASE_ANON_KEY · SUPABASE_SECRET_KEY · PGSSLMODE · WORK_DIR · ARCHIVE_RETENTION_DAYS · MIGRATIONS_ASSUME_APPLIED
Auth / session	SESSION_SECRET · FORM_UPLOAD_SECRET
AI	ANTHROPIC_API_KEY · OPENAI_API_KEY
Voice	TWILIO_ACCOUNT_SID · TWILIO_AUTH_TOKEN · TWILIO_PHONE_NUMBER · DEEPGRAM_API_KEY · ELEVENLABS_API_KEY · ELEVENLABS_AGENT_ID · ELEVENLABS_WEBHOOK_SECRET · VOICE_WEBHOOK_SECRET · VOICE_DEFAULT_ORG_ID · VOICE_SYSTEM_USER_ID · VOICE_DEFAULT_JOB_ID · LPC_TWILIO_NUMBER · LPC_VOICE_ORG_ID · LPC_ELEVENLABS_AGENT_ID · LPC_ELEVENLABS_WEBHOOK_SECRET · LPC_ELEVENLABS_WEBHOOK_SECRET_2 · PORT_VOICE_NOTIFICATION_EMAILS
Email	RESEND_API_KEY · EMAIL_REDIRECT_TO · CONTACT_TO_EMAIL
Microsoft	MICROSOFT_TENANT_ID · MICROSOFT_CLIENT_ID · MICROSOFT_CLIENT_SECRET · AZURE_TENANT_ID · AZURE_CLIENT_ID · AZURE_CLIENT_SECRET · SHAREPOINT_SITE_ID · SHAREPOINT_LIST_ID · SHAREPOINT_DRIVE_ID · SHAREPOINT_SITE_URL · TEAMS_WEBHOOK_URL

GROUP	NAMES
Maps (client)	VITE_LINZ_API_KEY · VITE_LINZ_DATA_KEY · VITE_MAPBOX_TOKEN
Infra / misc	APP_URL · NODE_ENV · PORT · LOG_LEVEL · ALLOWED_ORIGINS · FLY_MACHINE_ID · RC_ORG_ID · CI · PLAYWRIGHT_BASE_URL

Fail-closed at startup: `DATABASE_URL`, `SESSION_SECRET` only. Values are never recorded in this package.

F. Test inventory

FILE	COVERS
server/__tests__/ai-assistant.test.ts	Ask RC assistant behaviour
server/__tests__/ai-drone-analyzer.test.ts	Drone image analyzer
server/__tests__/ai-form-extractor.test.ts	Voice form field extraction
server/__tests__/analysis-sharepoint-sync.test.ts	Analysis → SharePoint sync service
server/__tests__/anchor-import-ai.test.ts	Anchor plan import AI parsing
server/routes/__tests__/drone-surveys.test.ts	Drone survey routes
server/lib/ai-json-repair.test.ts	AI JSON repair helper
server/lib/error-message.test.ts	Error message helper
e2e/smoke.spec.ts	Playwright smoke test — NOT RUNNABLE (@playwright/test undeclared)

CI (GitHub Actions): tsc · eslint --max-warnings 0 · vitest run. No e2e in CI.

G. Screenshot register

ZERO SCREENSHOTS

No screenshots were captured or committed: the V1 application was not run during this inspection (see section 00), and no production captures were available to sanitise. Any future screenshots added to `public/screenshots/` must be sanitised per the screenshot-safety rules and registered here.

H. Evidence register summary

50 register entries (EV-001 … EV-050) across platform, auth, tenancy, capture, testing, safety, voice, integrations, offline, deployment, security and legacy areas — each with classification, source file, confidence and validation column: `evidence/V1_Evidence_Register.csv`. Distribution: 33 IMPLEMENTED · 10 PARTIAL · 4 LEGACY · 1 DORMANT · 1 NOT IMPLEMENTED · 1 UNVERIFIED.

I. Legacy and dormant register

24 items (LD-01 … LD-24): dropped aviation/Co-Pilot artefacts, dormant pages and stubs, placeholder nav items, stale docs, committed one-off scripts, and the attached_assets directory. Full register: `evidence/V1_Legacy_and_Dormant_Register.csv`.

J. Known limitations of this package

- The application was not run; runtime behaviour, screenshots, live data volumes, and production configuration are all outside the evidence base.
- Client hook/component data-flow was traced at module level, not exhaustively per field.
- Production infrastructure (Fly machine list, Supabase settings, secret values) is invisible to the repository and therefore to this package.
- Counts (line numbers, occurrence counts) are as of commit `975a2ea` and will drift.

K. Glossary

TERM	MEANING
Anchor install log	The core geotech record: one row per anchor carrying drill, grout and status data (anchor_install_logs).
DAS	Daily Activity Sheet — per-shift crew and plant hours; "DAS specs" are the org rate/roster catalogues.
Drill Summary	The management review surface over anchor install logs (/anchor-logs).
Face photo	A rock-face photograph used as a spatial canvas for anchor pins and annotations.
Feature flag	Per-tenant boolean in tenant_config.featureFlags gating navigation items.
JSA / SWMS	Job Safety Analysis / Safe Work Method Statement — safety documents with daily sign-ins.
orgType	Organisation type enum (engineering, port, + dormant values) that shapes navigation.
PDS	Page Definition Sheet — the per-screen specification format used in section 05.
RLS	Postgres row-level security — enabled deny-by-default on all public tables.
RockControl	The V1 application's internal product name.
Tenant	A customer organisation; one row in organizations + tenant_config, served by a custom domain.
Voice intake	The Twilio → ElevenLabs → Claude pipeline that files forms from phone calls.

L. Repository baseline and methodology

- **Evidence source:** TACEDGE Geotech, branch `claude/tacedge-v1-product-definition-bxrl5e`, commit `975a2ea352f1e1ab260c622a66ab59291d8b81b1` (2026-07-05). Inspected 2026-07-17, read-only, working tree clean before and after.
- **Output repository:** tacedge-v1-PD, branch `claude/tacedge-v1-product-definition-bxrl5e`. Framework: Astro 5 static build with Pagefind search and Mermaid-rendered SVG diagrams.
- **Method:** parallel structured code inspection (routes, schema, migrations, services, client pages, configuration, docs, git history) consolidated into the evidence registers, then into this site. Counts were produced with repository-wide searches at the pinned commit.
- **Fonts:** the specified Play / Be Vietnam Pro / JetBrains Mono are self-hosted via the OFL-licensed @fontsource packages — no substitutions were needed.
- **Statistics:** 75 capabilities · 26 page definition sheets covering 42 registered routes · 62 active tables (+22 dropped) · ~230 API endpoints · 50 evidence entries · 24 legacy/dormant items · 34 open questions.

M. Confidence assessment

High confidence: repository structure, schema, routes, capabilities' existence, security posture, debt findings — all code-traceable. **Medium confidence:** behavioural descriptions of complex client surfaces, offline depth, notification consumption, storage privacy. **Low/unverified:** anything requiring runtime or production access —

live usage, reliability, AI accuracy, deployed machine topology. The package deliberately does not disguise these gaps; they are the first work items of the scoping sprint.